IMAGING PERFUSION RESTRICTIONS FROM EXTRACELLULAR SOLID STRESS
AN OPEN-LABEL LOSARTAN STUDY

Protocol Identification Number: ImPRESS-losartan study

EudraCT Number: 2018-003229-27

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- Norwegian REC approval number: **2018/1627**
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- Norwegian REC approval number: **2017/1875**
  Title: *ImPRESS - Redusert Perfusjon i Hjernekreft*
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SIGNATURE PAGE

Title	ImPRESS - Imaging Perfusion Restrictions from Extracellular Solid Stress An open-label losartan study
Protocol ID no:	ImPRESS-losartan study Version 7.0 dated 08 DEC 2024.
EudraCT no:	2018-003229-27

I hereby declare that I will conduct the study in compliance with the Protocol, ICH GCP and the applicable regulatory requirements:

Name	Title	Role	Signature	Date
Paulina B. Due-Tønnessen, MD	Head of Division. Division of Radiology and Nuclear Medicine , Oslo University Hospital	Sponsor		
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Inge Christoffer Olsen, PhD	Statistician, Research Support Services, Oslo University Hospital	Statistician		

SUMMARY OF CHANGES FROM PROTOCOL VERSION 6.0 TO VERSION 7.0

- Updated information of funding sources
- Updated information of ethical approvals
- Updated institutional information (new department name)
- Updated anticipated recruitment period
- Updated information on primary endpoints by specific descriptions of the analyses of collected MR images
- Updated information on secondary endpoints by specific descriptions of the analyses of collected blood samples
- Updated information on secondary endpoints by specific descriptions of the analyses of collected MR images
- Updated list of abbreviations and definitions of terms

SUMMARY OF CHANGES FROM PROTOCOL VERSION 5.0 TO VERSION 6.0

It has been necessary to update the protocol with regard to the rapidly changing treatment regimens for lung cancer patients with brain metastasis (Study B part of the protocol). The following changes have been implemented in Protocol version 6:

- Change of sub-investigator with main responsibility for study B subjects
- Inclusion criterium No. 8 has been modified to be valid for Study A (glioblastoma) only. There is no reason to require a previous history of stereotactic radiosurgery and immunotherapy treatment for Study B subjects in order to be included in the study
- Inclusion criterium No. 9 has been updated with the option of scheduled stereotactic radiosurgery for Study B subjects as this is a frequently used treatment in Standard of Care practice for brain metastasis.
- Exclusion criterium No. 19: "No previous history of immunotherapy at time of inclusion", has been removed for several reasons. Multiple immunotherapy agents have been approved for first line treatment in lung cancer. Often the subject will already have received immunotherapy before the subject is diagnosed with advanced disease and referred for further treatment. This criterium would have limited inclusion possibilities and in addition it was in conflict with previous inclusion criterium 8.
- Section 5.6 has been updated with regard to allowed time point for stereotactic radiosurgery and with the current systemic anti-neoplastic treatment.
- Section 6.6.2 has been updated with endocrinal blood tests at time of screening and during treatment. These blood tests are part of Standard of Care practice for lung cancer patients with brain metastasis receiving systemic anti-neoplastic treatment.
- Section 6.6.2 has been updated with a procedure for study procedures and Standard of Care treatment performed at other locations then the study center. This is particularly relevant for Study B subjects. Also the

pandemic situation has enforced us to work in another way in order not to lose subjects and study data, in addition to avoid protocol deviations.

- Appendix C – Trial Chart for Group B: Brain metastasis has been updated with regard to Standard of Care practice for group

- Appendix F: “Confidentiality Agreement” has been removed from the protocol. This document is revised regularly and latest version is always available in hospital internal document system (eHåndbok).

- Minor administrative changes of no significance to trial design, efficacy and safety other than mentioned above.

SUMMARY OF CHANGES FROM PROTOCOL VERSION 4.0 TO VERSION 5.0

The following changes have been implemented in Protocol version 5

- Change of representative for Sponsor to Paulina B Due-Tønnesen MD, Head of Radiology and Nuclear Medicine Department at Oslo University Hospital, Norway

- Updated information of funding sources

- Inclusion criterium No. 8 has been updated to include subjects with a previous history of a neurosurgical procedure (Study A only), or stereotactic radiosurgery and immunotherapy (Study B).

- Inclusion criterium No. 9 has been expanded for Study A – recurrent glioblastoma subjects to: Scheduled for chemotherapy and/or radiotherapy and/or stereotactic radiosurgery instead of only scheduled for chemotherapy

- Exclusion criterium No. 9 has been specified to be valid for recurrence of tumor inside the radiotherapy target volume. This allows subjects with recurrence of tumor outside the radiotherapy volume to participate in the study.

- Exclusion criterium No. 10 has been revised to be only valid for Study B subjects – brain metastasis.

- Exclusion criterium No. 16 has been specified to be valid for Study B subjects as only these subjects will be assessed for immunotherapy.

- New exclusion criterium No. 19 for Study B subjects. No previous history of immunotherapy at time of study inclusion is added as an exclusion criterium.

- Study A – both cohorts: Treatment with IMP for Study A subjects has been extended with 2 days before the second randomization. That means that in the first 3 Cycles of the study, the subjects can have either a minimum treatment period with IMP of 16 days (14 days before) or a maximum period of 44 days (42 days before).

- Section 5.6 has been updated with regard to Standard of Care Treatment for NSCLC according to National Guidelines
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- Sections 5.8 and 5.9 have been merged to Section 5.8 Compliance and Drug Accountability. Section has been updated with registration of IMP compliance and the criterium for exclusion of a subject when the IMP has not been taken to days in a row has been removed.

- Section 6.6.1 has been updated with information about the possibility to participate in another clinical imaging protocol with PET-CT imaging simultaneously to participation in ImPRESS.

- Section 9.1 has been updated with a change in electronic Data Capture System, using Viedoc software instead of EPI-data. The change in system also implies changes in section 9.5.1 Electronic CRF describing data collection and –management

- Appendix A – Trial Chart for Study A – Recurrent glioblastoma has been updated with one extra, exploratory blood sample at Dag 85 (±4 days). The reason for this is to compare MR biomarkers with the results of the research blood test at this time point.

- Appendix B and C – Trial Charts for Study A - Newly Diagnosed Glioblastomas and B - Brain Metastasis have been updated with regard to the baseline study specific MRI exam. The Trial Charts indicated that there should be a study specific MRI exam before surgery or SRS. This exam is of no value as a baseline for efficacy measurements and therefore the performance time point has been changed to the period 1-7 days before start of treatment (Day 1). This also allows us to obtain informed consent after surgery or SRS which is a better time point for the subject to decide if he/she wants to participate in this study.

- Appendix A, B and C – Trial Charts for both Study A cohorts and for Study B have been updated in order to improve the study logistics and to minimize number of study visits. In addition the Trial Charts have been adapted to hospital clinical routines in order to avoid deviation from protocol. The changes will not affect the safety surveillance.

- Minor administrative changes of no significance to trial design, efficacy and safety other than mentioned above

SUMMARY OF CHANGES FROM PROTOCOL VERSION 3.0 TO VERSION 4.0

The following changes have been implemented in Protocol version 4

- Inclusion criterium No. 3 has been updated to have measurable intracranial disease defined as at least one lesion that can be accurately measured in at least one dimension as ≥10 mm with MRI. The definition has been updated from ≥5mm to ≥10 mm to be compliant with RANO criteria for disease evaluation

- Section 5.3 Duration of Therapy has been updated. Study A subjects randomized to IMP dose 25mg and 50m will have a step-wedged treatment design where some patients will have extended IMP treatment with the same dose, while the others stop treatment at day 42. Appendix A and B Trial charts have been updated accordingly

- Section 7.1.1. Efficacy assessment of the primary endpoints has been updated with information on how to define the region of interest for the efficacy assessments of relative cerebral blood flow and solid stress. It is defined as a composite **primary target area** (**PTA**) including any contrast agent enhancing lesion or post-surgical cavity on the MR images (equaling the Gross Total Volume, GTV), as well as a 0.5 centimeter wide area surrounding the GTV, and while at all times within the Clinical Target Volume (CTV) if available (for **Study A – newly diagnosed glioblastoma** and **study B**). The 0.5 centimeter wide area surrounding the GTV represents the radiological definition of the peri-tumoral zone.

- Section 8.4 Time period for reporting AE and SAE has been updated with reporting periods for the different diagnosis groups.

- Section 8.8 Safety Committee has been updated with new members.

- Minor administrative changes of no significance to trial design, efficacy and safety other than mentioned above

PROTOCOL SYNOPSIS

Protocol title:	ImPRESS - Imaging Perfusion Restrictions from Extracellular Solid Stress An open-label losartan study
Sponsor:	Oslo University Hospital, Oslo, Norway
Phase and study type:	An open-label, single institutional phase II trial with an individual stepped-wedge, randomized, assessor-blinded, dose-finding design on three indications.
Investigational Medical Product (IMP):	Losartan (Cozaar ®)
Center(s):	Oslo University Hospital, Oslo, Norway
Study Period:	Estimated date of first patient enrolled: **01-NOV -2019** Anticipated recruitment period: **60 MONTHS** Estimated date of last patient included: **01-NOV-2024** End of trial defined as last patient-last visit. **01-MAY-2025**
Treatment Duration:	Expected treatment duration pr. patient: **Study A** (patients with recurrent glioblastoma) 168 days until and including cycle 12 of losartan (matching 6 cycles of chemotherapy). **Study A** (patients with newly diagnosed glioblastoma) 238 days until and including cycle 17 (matching 7 cycles of chemotherapy). **Study B** (lung cancer patients with metastases to the brain) 270 days until and including cycle 3 (matching 9 months of concomitant therapy)
Follow-up:	Expected follow-up period of IMP per participant: **Study A** (patients with recurrent glioblastoma) 168 days + 14 days **Study A** (patients with newly diagnosed glioblastoma) 238 days + 14 days **Study B** (lung cancer patients with metastases to the brain) 270 days + 14 days Follow up for Progression Free Survival and Overall Survival up to 24 Months
Objectives:	Primary study objectives: 1. To assess the dose-response relationship of the angiotensin II receptor inhibitor losartan on perfusion and mechanical solid stress in brain tumors. Key secondary objectives: 1. To assess the dose-response relationship of losartan on conventional and experimental radiographic characteristics in patients with glioblastoma and metastases to the brain

	2. To assess the dose-response relationship of losartan as add-on to standard treatment on clinical or patient reported endpoints in patients with glioblastoma and metastases to the brain 3. To assess the safety of losartan treatment in patients with glioblastoma and metastases to the brain
Endpoints:	Primary endpoints: **1.** Change from baseline in the radiographic biomarker relative cerebral blood flow (rCBF) [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study B** (lung patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7] **2.** Change from baseline in a radiographic biomarker representative of relative solid stress [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study B** (lung patients with metastases to the brain) Baseline, Day 90-14, Day 180-14, Day 270-14] Secondary endpoints: • Change from baseline in experimental radiographic biomarkers from MRI, including size of enhancing tumor, size of tumor associated edema, perfusion MRI, diffusion MRI, and/or MR Elastography [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 78-8/+1,. **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 148-8/+1, **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-14, Day 180-14, Day 270-14] • Drug tolerance of IMP (NCI-CTCAE v4.0) according to follow-up period, including number and frequency of losartan treatment emerging adverse events (AE), vital signs (blood pressure, pulse) and conventional laboratory blood parameters [Time Frame: **Study A** (patients with recurrent glioblastoma) 168 days + 14 days **Study A** (patients with newly diagnosed glioblastoma) 238 days + 14 days **Study B** (lung cancer patients with metastases to the brain) 270 days + 14 days] • Change from baseline in neurologic performance scores by Karnofsky Performance Score (KPS), Eastern Cooperative Oncology Group (ECOG) and/or Neurologic Assessment in Neuro-Oncology (NANO) scores. [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 42-44, Day 85±4, Day 169±4.

	Study A (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 42-44, Day 71±4, Day 155±4, Day 239±4. **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7] • Best overall response of radiographic status on MRIs of intracranial disease or corresponding treatment response by the Response Assessment in Neuro-Oncology (RANO) criteria (occurrence of radionecrosis, pseudoprogression or tumor progression). [Time Frame: **Study A** (patients with recurrent glioblastoma) Day 78±4, Day 162±4, **Study A** (patients with newly diagnosed glioblastoma) Day 148±4, Day 232±4, **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-14, Day 180-14, Day 270-14, Day 360±7.] • Total dose and change in steroids dosage during study treatment duration. [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 57±4, Day 85±4, Day 113±4, Day 127±4 for Lomustine and Day 141±4 for Temodal, Day 169±4. **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 71±4, Day 99±4, Day 127±4, Day 155±4, Day 183±4, Day 211±4, Day 239±4 **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7] • Change from baseline in the European Organization for Research and Treatment of Cancer (EORTC) quality of life questionnaire (QLQ BN20) core 30 version 3.0. [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 42-44, 169±4. **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 42-44, Day 155±4, Day 239±4. **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7] • Change from baseline in experimental biomarkers from blood and serum, including vascular endothelial growth factors (VEGF-A, VEGF-C, VEGFR2, CD31, FGF, PDGF, IGFBP7), angiopoietins (ANG-1, ANG-2, ANGPTL4), placental growth factors (PIGF), epidermal growth factor receptors (EGF, TGF-alpha), hyaluronan levels, collagens levels (collagen IV alpha 1, MMP-10, Tenascin C, CD163, CA9, THBS2), and inflammatory cytokines (cxcl9-11, CCL2, CX3CL1, ErbB2, Fas Ligand/TNFSF6, GITR/TNFRSF18, TNF-alpha, Granzyme B, IFN-gamma, IL-1beta, IL-2, IL-6, IL-8, IL-12). [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 42-44, Day 90±4, Day 155±4. **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 42-44, Day 225±4. **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7] • Six-months progression free survival (6M-PFS) [Time Frame: Day +180+7] • Progression free survival (PFS) [Time Frame: up to 24 months ±7 days]

	• Overall survival (OS) at 12±7 days and 24±7 days months [Time Frame: at 12 months, 24 months] • Overall survival [Time Frame: up to 24 months]
Study Design:	An individual stepped-wedge, randomized, assessor-blinded, dose-finding single-center trial on three indications.
Main Inclusion Criteria:	**1.** A histologically confirmed intracranial glioblastoma, WHO grade 4 (**Study A**), or a minimum of one radiographically confirmed metastasis to the brain from a primary non-small-cell lung cancer (**Study B**) **2.** Ability to undergo an MRI exam, including administration of a standard clinical dose of an MRI-specific contrast agent of gadolinium or similar **3.** Measurable intracranial disease (**Study A** – recurrent glioblastoma, and **Study B** only), defined as at least one lesion that can be accurately measured in at least one dimension as ≥10 mm with MRI **4.** Age ≥18 years **5.** Eligible for administration of the active substance (losartan) in concordance with study protocol, the criteria of the product label (Cozaar) and deemed fit for trial by the treating physician. **6.** An ECOG performance status of ≤2 or equivalent KPS of ≥60% **7.** Life expectancy from start of treatment of more than 3 months **8.** Previous history of neurosurgical procedure at time of study inclusion (**Study A** – only) **9.** Scheduled for chemotherapy and/or radiotherapy and/or stereotactic radiosurgery (**Study A** – recurrent glioblastoma), neurosurgery, radiotherapy and chemotherapy (**Study A** – newly diagnosed glioblastoma), immunotherapy and/or chemotherapy and/or stereotactic radiosurgery (**Study B**) **10.** Pre-study documentation on O^6-methylguanin-DNA-methyltransferase (MGMT) promoter methylation status and on the isocitrate dehydrogenase (IDH) gene mutation status of their disease (**study A** only) **11.** Organ functions of sufficient quality and robustness to undergo study treatment as determined by the study principal investigator (PI) or designee **12.** Female patients of childbearing potential (postmenarcheal, not postmenopausal (>12 continuous months of amenorrhea with no identified cause other than menopause), and no surgical sterilization) should use highly effective contraception and take active measures to avoid pregnancy while undergoing IMP treatment and for at least 14 days after the last dose. Birth control methods considered to be highly effective include combined (estrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation (oral, intravaginal or transdermal), progestogen-only hormonal contraception associated with inhibition of ovulation (oral, injectable or implantable), intrauterine device, intrauterine hormone-releasing system, bilateral tubal occlusion, vasectomized partner, sexual abstinence when it is the preferred and usual lifestyle of the subject. **13.** Ability to understand and the willingness to sign a written informed consent document
Main Exclusion Criteria	**1.** Hypersensitivity to the active substance (losartan) or to any of the excipients **2.** Patients on antihypertensive agents that cannot be substituted with losartan.

	3. Patients on medication that may induce hypotension and/or increase potassium levels and/or cause metabolism-related pharmacokinetic drug-drug interactions with losartan. If medication with such effects can safely be discontinued or replaced, the patient can be included in the study after a washout period before baseline. **4.** Patients with hepatic or renal impairment of any reason **5.** Patients with symptomatic hypotension of any reason **6.** Patients with primary hyperaldosteronism **7.** Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucosegalactose malabsorption should not take this medicine **8.** Inadequate recovery from toxicity and/or complications of previous therapy as determined by the treating physician **9.** Patients with evidence of recurrence inside the radiotherapy target volume less than 3 months since last radiotherapy fraction (**Study A** – recurrent glioblastoma only) **10.** For **Study B** subjects only. A diagnosis of immunodeficiency or hypersensitivity to PD-1 inhibitors or any excipients **11.** Uncontrolled intercurrent illness including, but not limited to, ongoing or active infection, symptomatic congestive heart failure, coronary heart disease and cerebrovascular disease, unstable angina pectoris, cardiac arrhythmia, angioedema, intravascular volume depletion, or psychiatric illness/social situations that would limit compliance with study requirements as determined by treating the physician **12.** Patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy **13.** Patient with known psychiatric or substance abuse disorders that would interfere with cooperation with the requirements of the study as determined by the treating physician **14.** Pregnant or breastfeeding patient **15.** Known additional active non-study related malignancy **16.** Applies to **Study B** patients qualifying for immunotherapy only: Active autoimmune disease that has required systemic treatment in the last 2 years (including use of non-study related disease modifying agents, corticosteroids or immunosuppressive drugs). Replacement therapy (e.g., thyroxine, insulin, or physiologic corticosteroid replacement therapy for adrenal or pituitary insufficiency, etc.) is allowed **17.** Known history of Human Immunodeficiency Virus (HIV) (HIV 1/2 antibodies) **18.** Unable to undergo brain MRI according to study protocol
Sample Size:	**153 participants in total.** The combined number of evaluable patients for the ***ImPRESS-losartan*** studies is estimated at 126 adult patients in order to achieve 45 evaluable patients in each of the **Study A** subcategories (recurrent and newly diagnosed glioblastoma) and 36 evaluable patients in **Study B** (brain metastases). To reach this number, including a buffer for study drop-out, we aim to include a total of **153 adult patients** (Study A: 54 + 54 and Study B: 45) in study circulation.

Efficacy Assessments:	**For the primary endpoints:** Tissue rCBF is derived from the Zierler's area-height relationship of the perfusion MRI-based tissue-concentration curve following an intravascular contrast agent bolus administration, and normalized to healthy-appearing reference. A proxy for tissue solid stress is derived from MR Elastography by estimating the tissue stiffness from the shear wave modulus G* and normalized to healthy-appearing reference. **For the secondary endpoints:** - From perfusion MRI using the parameters relative blood volume, mean transit time, and tissue extravasation following an intravascular contrast agent bolus administration, and normalized to healthy-appearing reference. - From vessel architectural imaging using the parameters vessel size index, relative mean vessel density, relative mean vessel caliber, corrected vortex area, direction map following an intravascular contrast agent bolus administration, and normalized to healthy-appearing reference. - From diffusion MRI using the parameters apparent diffusion coefficient, fractional anisotropy, cellular index - From MR Elastography using the parameters stiffness |G*|, shear phase angle Y, elasticity (shear storage modulus), viscosity (shear loss modulus) - From T1-weighted MRI using the total volume of enhancing tumor following an intravascular contrast agent bolus administration - From T2-weighted FLAIR MRI using the total volume of tumor edema - Drug tolerance by reported serious adverse events - Neurological performance by Karnofsky and/or ECOG and/or NANO scores - Radiographic best response by RANO (radionecrosis, pseudoprogression or tumor progression) - Steroid dosage and use - Blood pressure - Quality of life by use of the EORTC QLQ BN20 and QLQ-C30 forms - From blood sampling using Luminex Human Discovery Assay kits for vascular endothelial growth factors (VEGF-A, VEGF-C, VEGFR2, CD31, FGF, PDGF, IGFBP7), angiopoietins (ANG-1, ANG-2, ANGPTL4), placental growth factors (PIGF), epidermal growth factor receptors (EGF, TGF-alpha), hyaluronan levels, collagens levels (collagen IV alpha 1, MMP-10, Tenascin C, CD163, CA9, THBS2), and inflammatory cytokines (cxcl9-11, CCL2, CX3CL1, ErbB2, Fas Ligand/TNFSF6, GITR/TNFRSF18, TNF-alpha, Granzyme B, IFN-gamma, IL-1beta, IL-2, IL-6, IL-8, IL-12). - Hyaluronan from hyaluronan synthases by PCR - Clinical/radiographic 6-month PFS or median PFS

	- Patient survival at readout time points 12 and 24 months and/or median OS
Safety Assessments:	The safety profiles of Cozaar and concomitant medications are well known, and the focus is therefore on any unfavorable interactions with Cozaar and the standard-of-care treatment. Any adverse events with regard to a dose-response are also assessed. All adverse events (AE) will be recorded, but only AE of grade 3 (severe) or grade 4 (life-threatening) will be reported in the Case Report Form (CRF) and monitored for status of SAE (serious adverse events or SUSAR (suspected unexpected serious adverse reactions) by the study PI. This trial is open-label; therefore, the participant, the trial site personnel, the Sponsor and/or designee are not blinded to treatment. Drug identity (name, strength) is included in the label text. Storage, handling and preparation requirements will be handled in concordance with the Pharmacy's Standard Operating Procedures (SOP).
Other Assessments:	To further develop magnetic resonance imaging methods to identify personalized treatment options for patients with glioblastomas and metastases to the brain Per signed informed consent, collected study data (clinical, imaging and blood/serum) will be stored for up to 15 years or according to applicable regulation and further analyzed to address scientific questions and/or development of biological tests related to the IMP, concomitant therapy and/or cancer. The decision to perform any exploratory biomarker research studies on pseudonymous or anonymized data would be subject to a new scientific and ethical review and based on outcome data from this study or from new scientific findings related to the drug class or disease, as well as reagent and assay availability. In addition to the biomarkers specified in this chapter, exploratory anonymized biomarker research may be conducted on any radiographic, blood/serum or clinical data collected as part of the study. These studies would extend the search for other potential biomarkers relevant to the effects of the drugs given in combination in this study, and/or prediction of these effects, and/or resistance to the treatment, and/or safety, and these additional investigations would be dependent upon clinical outcome and data availability. As an example of such an exploratory analysis, the RPI signature (Restricted Perfusion Imaging, distribution of image voxels) of every patient's tumor will be assessed by combining MRI-based measures of vascular efficiency (relative mean vessel caliber multiplied by blood flow) from MR perfusion with those of tissue solid stress from MR Elastography (~stiffness). The RPI signature will be compared over time using between-exam longitudinal analyses for treatment response. Here, the magnitude of displacement (per axis value) of the centroid of the image voxels in (x,y) coordinates as presented in the scatter plot will constitute a quantitative measure of treatment chance. Any centroid shift equivalent of a rightwise (x-axis) and/or downward (y-axis) movement between two exams will constitute a positive change. A significant change in the treatment group compared to baseline value and/or a non- or a different-treatment group is regarded as a treatment-induced change. Another example of an exploratory analysis includes measures of tissue solid stress derived from MR Elastography by estimating the stiffness from the shear wave modulus G* multiplied by the apparent tissue displacement from Jacobean maps, and normalized to healthy-appearing reference

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LIST OF ABBREVIATIONS AND DEFINITIONS OF TERMS

Abbreviation or special term	Explanation
2D	2-Dimensional
3D	3-Dimensional
ACE	Angiotensin Converting Enzyme
ADC	Apparent Diffusion Coefficient
AE	Adverse Event
ALT	Alanine aminotransferase
ARB	Angiotensin-receptor blocker
ANG-1	Angiotensin II receptor type 1
ANG-2	Angiotensin II receptor type 2
ANGPTL4	Angiopoietin-related protein 4
ATC	Anatomical Therapeutic Chemical Classification System
AUC	Area under the curve
CA	Carbonic anhydrase
CCL2	C-C motif ligand 2
CI	Cellular index
COX-2	Cyclooxygenase-2
CD31 / CD 163	Cluster of differentiation 31 / 163
CR	Complete Response
CRF	Case Report Form (electronic/paper)
CSA	Clinical Study Agreement
CSR	Clinical Study Report
CT	Computer Tomography
CTC	Common Toxicity Criteria (for cancer trials only)
CTCAE	Common Terminology Criteria for Adverse Event (for cancer trials only)
CTV	Clinical Target Volume
CYP2C9	Cytochrome P450 2C9
CX3CL1	Fractalkine
DAE	Discontinuation due to Adverse Event
EC	Ethics Committee, synonymous to Institutional Review Board (IRB) and Independent Ethics Committee (IEC)
ECOG	Eastern Cooperative Oncology Group
EGFR	Epidermal Growth Factor Receptors
FLAIR	Fluid-attenuated inversion recovery
FA	Fractional Anisotropy
GCP	Good Clinical Practice
GTV	Gross Total Volume
HIV	Human Immunodeficiency Virus
IB	Investigator's Brochure
ICF	Informed Consent Form
ICH	International Conference on Harmonization
IGFBP	Insulin-like growth factor-binding protein
IL	Interleukin
IMP	Investigational Medicinal Product (includes active comparator and placebo)
INF	Interferon
KPS	Karnofsky Performance Status
K^{trans}	Surface area product (indicator of capillary permeability)
MAR	Missing At Random
MCP-Mod	Multiple Comparison Procedure and Modelling
MMP	Matrix metalloproteinase
MR	Magnetic Resonance
MRE	Magnetic Resonance Elastography
MRI	Magnetic Resonance Imaging

Abbreviation or special term	Explanation
MTT	Mean Transit Time
nLIC	Non-Linear Image Co-registration
NSAID	Nonsteroidal anti-inflammatory drug
NSCLC	Non-Small Cell Lung Cancer
OS	Overall survival
OUH	Oslo University Hospital
PACS	Picture Archiving and Communication System
PD	Progressive Disease
PD-1	Programmed cell death protein 1
PDL1	Programmed death-ligand 1
PEGPH20	Pegylated recombinant human hyaluronidase
PET	Positron Emission Tomography
PFS	Progression Free Survival
PHI	Protected Health Information
PI	Principal Investigator
PIGF	Placental growth factors
PR	Partial Response
PRA	Plasma Renin Activity
PTA	Primary Target Area
QoL	Quality of Life
rCBF	Relative Cerebral Blood Flow
rCBV	Relative Cerebral Blood Volume
RANO	Response Assessment in Neuro-Oncology
RPI	Restricted Perfusion Imaging
RSI	Restricted Spectrum imaging
RTOG	Radiation Therapy Oncology Group
SAE	Serious Adverse Event
SAP	Statistical Analysis Plan
SD	Stable Disease
SHG	Second-harmonic generation
SmPC	Summary of Product Characteristics
SOP	Standard Operating Procedure
SRS	Stereotactic Radiosurgery
T1	Longitudinal relaxation time
T2	Transversal relaxation time
TGF	Transforming growth factor
THBS2	Thrombospondin-2
TNFRSF	Tumor necrosis factor receptor superfamily
TGF	Transforming growth factor
USB	Universal Serial Bus
VAI	Vessel Architectural Imaging
VEGF	Vascular Endothelial Growth Factor
VEGFR	Vascular Endothelial Growth Factor and Its Receptor
VSI	Vessel Size Index

1 INTRODUCTION

1.1 Background - solid stress in cancer

The topological and structural heterogeneity of tumor microcirculation is a decisive determinant for disease progression and the potential for treatment response[1-4]. Unlike the balanced circulatory arrangements in normal tissue, the tumor bed is characterized by a chaotic network of disorganized, permeable, and tortuous vessels[5,6]. Attributed to a complex interplay between physical forces and the imbalance of pro- and anti-angiogenic factors, tumor-associated vessels have a wide spectrum of calibers, from small and compressed capillaries to abnormal and dilated veins[7-12]. While the processes of neo-angiogenic pathways and anti-angiogenic therapies are among the most-studied facets of cancer[13-17], only a limited number of pre-clinical studies pursue the hypothesis that a mechanical force caused by proliferating tumor cells and the fibrous extracellular matrix can squeeze fragile cancer vessels shut and prevent drug delivery[18-21]. This is in contrast to another and better understood force, the interstitial fluid pressure, which is upregulated in cancer because of leaky blood vessels and poor clearance by impaired lymphatic systems[22]. Although high interstitial fluid pressure is associated with poor prognosis in a range of cancers because of reduced extra-vascular drug transportation and increased tissue hypoxia[22-24], its pressure level equals that of microvascular pressure (~10mmHg)[19] and therefore cannot induce compression of permeable tumor vessels[20,25]. Instead, a porous vessel is easily crushed by applying an external, non-fluid, mechanical force - and in tumors, such a force is in abundance; the solid stress of proliferating cells and the extra-cellular matrix[25].

The connective tissue framework in a solid tumor, the stroma, surrounds the cancer cells and consists of unspecialized stem cells and the extracellular matrix[26]. The major constituents of the matrix are long, filamentous fibers of proteins that resist tension and non-fibrous polysaccharides that resist compression. Together, these components strengthen tissue structures, provide means of cellular transportation or movement and hydrate the matrix. In cancer, the extracellular matrix is remodeled (desmoplasia) by upregulated levels of collagen proteins (up to 90% of the matrix)[27] and hyaluronan, among others, that induce tissue stiffness and enables proliferating cells to generate tractional forces on their surroundings[28,29]. Tumor vessels which lack the complete coverage by pericytes and basement membrane are structurally weak and cannot resist these forces, resulting in vascular compression and reduced blood flow[25,30,31].

The cancerous microenvironment caused by impaired perfusion is associated with hypoxia and low pH[32], as well as reduced penetration of anti-cancer drugs[33,34] - all factors that contribute to tumor progression and treatment resistance[32,35]. Hypoxic cancer tissue is difficult to treat for several reasons; first, an acidic microenvironment attenuates the killing potential of immune effector cells and reprograms the resident macrophages (phagocytes) in to a pro-tumorigenic and immuno-suppressive phenotype[19,36]. Second, hypoxia may select for more malignant cells because cells that respond to physiological cues normally undergo apoptosis under hypoxic conditions[32]. Third, intermittent hypoxia causes radio-therapeutic resistance[32,35].

We hypothesize that measuring how mechanical forces of the tumor microenvironment restrict perfusion and promotes treatment resistance will be critical for patient care. ***ImPRESS-losartan*** focus on brain cancers, a patient group with dismal prognosis and a cancer type where the proposed studies are expected to have especially appealing effects.

1.2 Background - treatment options that may target solid stress in cancer

Current standard-of-care for patient with brain tumor in Norway includes chemotherapy, radiotherapy and/or surgery, depending on diagnosis and extent of disease. Although the majority of cancers eventually relapse, these treatment options have been somewhat successful in the sense that treated patients live longer than untreated, but often experience clinical and neurological deficits[37-42]. The heterogeneity within each patient group is high, both in terms of response to treatment and time to progression, and combined with the often rapid relapse and growth of these cancers, diagnostic assessment and therapy response monitoring are particularly important. Surgical debulking of tissue is an effective tool for reducing solid stress on a macroscopic level. This treatment option however, is not available readily for repeated use, nor does it target the components of the extracellular matrix. A simple, yet unspecific approach for relieving solid stress is by radiation-induced killing of radiosensitive cells that obstruct nearby vessels[35]. Because this approach favors oxygen-rich microenvironments, it is best used as a second-line defense.

The importance of intact immune surveillance in controlling outgrowth of neoplastic transformation has been known for decades. Immunotherapy is a promising therapeutic option that is now finally coming of age with astonishing results of prolonged patient survival, especially for malignant melanomas[27]. Immunotherapeutic antibodies work by blocking inhibitory receptors on lymphocytes to overcome immune tolerance and a wide range of therapeutic options are currently in clinical trials[43,44]. Among these, *ipilimumab*, an antibody targeting the inhibitory T-cell costimulatory molecule CTLA-4, is already approved for patients with metastatic melanomas. Moreover, *nivolumab*, *pembrolizumab* and *atezolizumab* are potent and highly selective humanized monoclonal antibodies designed to directly block the interaction between *programmed cell death protein 1* (PD-1) and its ligands, PD-L1 and PD-L2. The interaction between the PD-1 and PD-L1 is hijacked by tumors to suppress the immune system and prevent activation of an immune response. Preclinical data demonstrates infiltration of activated regulated T-cells, and PD-L1 expression in brain metastases of various histologies[45]. Given the resulting nonspecific immune activation, this therapeutic approach is appealing for a wide variety of tumor types as it is not dependent on tumor expression of specific molecules that is critical for targeted therapy (e.x. *erlotinib* for EGFR mutant lung cancer). In fact, immunotherapy may be more effective in tumors with more antigen expression or heterogeneity as can be measured with imaging[46]. The PD-L1 inhibitors *pembrolizumab and atezolizumab* are currently approved in Norway for the treatment of patients with non-small cell lung cancer (NSCLC), including those with advanced metastatic disease. Overall, PD-L1 inhibitors are well tolerated with fewer side effects than ipilimumab.

Alternative strategies that directly target components of the extracellular matrix, in particular collagen and hyaluronan, have also been suggested[25]. Of these, pegylated recombinant human hyaluronidase (PEGPH20), an inhibitor of hyaluronan, reached early-phase trials in patients with pancreatic cancers. Together with chemotherapy this approach resulted in a doubled progression-free survival compared to patients on chemotherapy alone (9.3 vs. 4.3months)[47,48]. However, no significant survival benefits were observed and because of the highly aggressive enzymatic depletion by targeted drugs like PEGPH20, this treatment option is no longer advised based on patients experiencing a range of adverse events, including peripheral edema, muscle spasms, neutropenia, myalgia, fatigue and extremity pain[47,48].

A widely prescribed and safe class of medicines with decades of use against hypertension, angiotensin-receptor blockers (ARBs), that acts specifically on the renin-angiotensin system, may provide an alternative mechanism for modulating solid stress. In non-brain cancer models, ARBs are found to not only lower blood pressure, but also decompress vessels by impairing collagen synthesis[49] and depleting hyaluronan[50] through downstream inhibition of transforming growth factor β1 (TGF-β1). In the brain, renin, a protein which produces vascular growth factor angiotensin (Ang) I, is expressed in astrocytes whereupon enzymes convert AngI to the active AngII - both factors essential to glioma progression [51-54]. In brain cancer patients, because there is a gap in knowledge and no available data, we do not know to what effect ARBs decompress vessels. However, ARBs inhibit AngI upstream of TGF-β1, which is highly expressed in brain cancers[55,56], and thus constitute a potential target for treatment if we can measure its effects on solid stress and the vasculature.

1.3 Clinical Experience with Investigational Medicinal Product (IMP)

We will use losartan (Cozaar®) as the angiotensin-II inhibitor, an affordable and widely prescribed medicine. Cozaar® has marketing authorization in the participating country (Norway) and is available for clinical use at Oslo University Hospital. After decades of clinical use per label use, losartan is considered very safe and with few adverse effects compared to strategies using enzymatic depletion (i.e. pegylated recombinant human hyaluronidase), TGF-β1 inhibition[57], or even angiotensin converting enzyme (ACE) inhibitors[58,59]. To this end, renin–angiotensin system inhibitors are presented as nontoxic and usually considered active for reducing blood pressure in hypertensive patients only, while producing no or limited adverse effects in healthy individuals[60].

Per label instructions, Losartan is a synthetic oral angiotensin-II receptor (type AT1) antagonist. Angiotensin II, a potent vasoconstrictor, is the primary active hormone of the renin/angiotensin system and acts specifically by inducing collagen production and hyaluronan synthesis. Angiotensin II binds to the AT1 receptor found in many tissues (e.g. vascular smooth muscle, adrenal gland, kidneys and the heart) and elicits several important biological actions, including vasoconstriction and the release of aldosterone. Angiotensin II also stimulates smooth muscle cell proliferation. Losartan selectively blocks the AT1 receptor. In vitro and in vivo losartan and its pharmacologically active carboxylic acid metabolite E-3174 block all physiologically relevant actions of angiotensin II, regardless of the source or route of its synthesis. Losartan does not have an agonist effect nor does it block other hormone receptors or ion channels important in cardiovascular regulation. Furthermore, losartan does not inhibit ACE (kinase II), the enzyme that degrades bradykinin. Consequently, there is no potentiation of undesirable bradykinin-mediated effects. During administration of losartan, removal of the angiotensin II negative feedback on renin secretion leads to increased plasma renin activity (PRA). Increase in the PRA leads to an increase in angiotensin II in plasma. Despite these increases, antihypertensive activity and suppression of plasma aldosterone concentration are maintained, indicating effective angiotensin II receptor blockade. After discontinuation of losartan, PRA and angiotensin II values fell within three days to the baseline values. Both losartan and its principal active metabolite have a far greater affinity for the AT1-receptor than for the AT2-receptor. The active metabolite is 10-40 times more active than losartan on a weight for weight basis.

Therapeutic label indications include treatment of; essential hypertension in adults and children and adolescents 6-18 years of age; renal disease in adult patients with hypertension and type 2 diabetes; and chronic heart failure in adult patients when treatment with ACE inhibitors is not considered suitable due to incompatibility.
Please refer the Summary of Product Characteristics (SmPC) for complete information on Cozaar® label instructions (Cozaar at https://www.legemiddelsok.no/).

Following oral administration, losartan is well absorbed and undergoes first-pass metabolism, forming an active carboxylic acid metabolite and other inactive metabolites. The systemic bioavailability of losartan tablets is approximately 33%. Mean peak concentrations of losartan and its active metabolite are reached in 1 hour and in 3-4 hours, respectively. Both losartan and its active metabolite are ≥99% bound to plasma proteins, primarily albumin. The volume of distribution of losartan is 34 liters. About 14% of an orally-administered dose of losartan is converted to its active metabolite. Following oral and intravenous administration of 14C-labelled losartan potassium, circulating plasma radioactivity primarily is attributed to losartan and its active metabolite. Minimal conversion of losartan to its active metabolite was seen in about one percent of individuals studied. In addition to the active metabolite, inactive metabolites are formed. Plasma clearance of losartan and its active metabolite is about 600 ml/min and 50 ml/min, respectively. Renal clearance of losartan and its active metabolite is about 74 ml/min and 26 ml/min, respectively. When losartan is administered orally, about 4% of the dose is excreted unchanged in the urine, and about 6% of the dose is excreted in the urine as active metabolite. **The pharmacokinetics of losartan and its active metabolite are linear with oral losartan potassium doses up to 200 mg. Following oral administration, plasma concentrations of losartan and its active metabolite decline polyexponentially, with a terminal half-life of about 2 hours and 6-9 hours, respectively.** During once-daily dosing with 100 mg, neither losartan nor its active metabolite accumulates significantly in plasma. Both biliary and urinary excretions contribute to the elimination of losartan and its metabolites. Following an oral dose administration of 14C-labelled losartan in man, about 35% of radioactivity is recovered in the urine and 58% in the feces. In elderly hypertensive patients the plasma concentrations of losartan and its active metabolite do not differ essentially from those found in young hypertensive patients. In female hypertensive patients the plasma levels of losartan were up to twice as high as in male hypertensive patients, while the plasma levels of the active metabolite did not differ between men and women. In patients with mild to moderate alcohol-induced hepatic cirrhosis, the plasma levels of losartan and its active metabolite after oral administration were respectively 5 and 1.7 times higher than in young male volunteers. Plasma concentrations of losartan are not altered in patients with a creatinine clearance above 10 ml/minute. Compared to patients with normal renal function, the area under the plasma drug concentration-time curve (AUC) for losartan is about 2-times higher in hemodialysis patients. The plasma concentrations of the active metabolite are not altered in patients with renal impairment or in hemodialysis patients. Neither losartan nor the active metabolite can be removed by hemodialysis.

Losartan in cancer:

In pre-clinical breast and pancreatic cancer models, losartan is shown to reduce stromal collagen and hyaluronan production, thereby actively alleviate solid stress and resulting in increased vascular perfusion and reduced tissue hypoxia[50]. In humans, recent meta-analyses[60,61] suggest that use of renin–angiotensin system inhibitors (including ARBs) significantly prolong survival in patients with renal cell carcinoma, gastric cancer, pancreatic cancer, hepatocellular carcinoma, upper-tract urothelial carcinoma, and bladder cancer. A trend towards better outcome was observed in rectal/colorectal cancer, lung cancer, prostate cancer, glioblastoma, head and neck squamous cell carcinoma, oropharynx cancer, and melanoma. Conversely, the RAS inhibitors showed negative effects in patients with acute myelocytic leukemia or multiple myeloma[60]. Moreover, for patients with glioblastoma in particular, the added value in terms of overall survival is debated. A large retrospective study in over 1300 glioma patients suggested renin–angiotensin system inhibitors were associated with prolonged survival[62]. However, although well tolerated, another recent study on losartan found no added survival compared to placebo in unselected patients with glioblastomas[63]. Although limited studies focused on the side effects of renin–angiotensin system inhibitors in cancer patients, Keizman et al reported that no inadvertent interactions were observed in patients receiving RAS inhibitors concurrently with sunitinib[64]. A recent study by Johansen et al in 243 patients with glioblastomas reported tendencies towards prolonged PFS and OS, as well as no added toxicity in a subgroup of patients (N=26) who received renin–angiotensin system inhibitors (including losartan) in combination with bevacizumab[65]. Moreover, there is an overwhelming body of evidence for the cardioprotection afforded by renin–angiotensin system inhibitor treatment[66].

Currently, there are six human cancer studies involving losartan registered on clinicaltrials.gov (search key words: cancer AND losartan). Of these, three studies actively pursue the role of losartan as an experimental anti-cancer drug. Two studies use losartan for adult patients with pancreatic cancer in combination with the FOLFIRINOX cocktail, as well as with- (ClinicalTrials.gov Identifier: NCT03563248, Status: recruiting), or without (NCT01821729, Status: active, not recruiting) the immunotherapeutic drug nivolumab. Both studies use losartan as an off-label dug by including patients without high blood pressure up-front, and as identified by a lower inclusion limit of a baseline systolic blood pressure above 100 mm Hg. The third study (ASTER trial; NCT01805453, Status: active, not recruiting), is a randomized, placebo-controlled trial to assess whether or not the addition of Losartan to standard of care can reduce steroid requirement during radiotherapy in patients with newly diagnosed glioblastoma. Preliminary findings are reported[63], where patients with a histologically confirmed newly diagnosed glioblastoma after biopsy or partial surgical resection were randomized between Losartan (n=37 patients) or placebo (n=38 patients) in addition to standard of care by radiotherapy and temozolomide. The primary objective was to investigate the steroid dosage required to control brain edema on the last day of radiotherapy in each arm. The secondary outcomes were steroid dosage one month after the end of radiotherapy, assessment of cerebral edema on MRI, tolerance, and survival. No difference in the steroid dosage required to control brain edema on the last day of radiotherapy, or one month after completion of radiotherapy, was seen between treatment arms. A trend towards reduction of peritumoral edema over time was seen on MRI, but this reduction did not reach statistical significance. Median overall survival was similar in both arms. No assessments of tissue solid stress or perfusion were performed. The potential survival benefits as reported by this research group however, are conflicting. One other study showed no apparent survival benefit[67], whereas another, small retrospective study by the same group *did* show prolonged overall survival (+3.8months)[54].

1.4 Rationale for the Study and Purpose

While a new therapeutic drug may target a specific pathway with great success, most cancer treatments do not account for other components of the microenvironment that also regulate tumorigenesis[19,27,68]. We hypothesize that measuring how mechanical forces of the tumor microenvironment restrict perfusion and promotes treatment resistance will be critical for patient care. The ***ImPRESS-losartan*** study focus on brain cancers, a patient group with dismal prognosis and a cancer type where the proposed studies are expected to have especially appealing effects. Moving from the current targeted-drug-delivery approach to a make-drugs-reach-their-target approach constitutes a potential game-changer in the way we assess cancer treatment. There are no available diagnostic strategies on how to measure impaired vasculature from solid stress in human cancers. ***ImPRESS-losartan*** will reveal the mechanisms of this mechanical force and its treatment options in human cancers by going beyond state-of-the-art and introduce a new paradigm for clinical cancer imaging. Because compressive growth-induced stress in human tumors (35-142mmHg) is much higher than in animals (2-60mmHg)[25], a pre-clinical analog is not sufficient to reach our objectives.

Hypothesis - losartan works in patients with impaired tumor tissue:

In addition to its availability, safety and long history of clinical experience, ARBs are promising as anti-cancer therapy because they target all stromal components including cancer-associated fibroblasts, hyaluronic acid and collagen[50]. Moreover, losartan is a particularly interesting agent for cancer because of its high drug penetration[69,70] with a polar surface area of 78Å^2 for its active ingredient (losartan potassium) versus 90Å^2 for the blood-brain barrier (and much larger pores in permeable brain tumors). **However, for losartan to have any clinical impact, the tissue of interest must be impaired in such a way that the administration of losartan effectively alleviates upregulated solid stress and improve perfusion. Correspondingly, a tissue without these properties will probably have little or no added benefit of losartan, adding to the importance of the MRI-based baseline stratification of tumor status.** In a recent study of 64 pre-surgery adult patients with glioblastomas[71], our group retrospectively quantified regional differences in perfusion and found that 52% (n=33) of the patients had reduced perfusion over the distance of up to 12 mm away from the tumor edge. For patients with brain metastases from breast cancer, a collective reduced perfusion was observed in 34 patients[71]. These proposed mechanisms of action for losartan contrasts those of the ASTER trial, where losartan was believed to reduce vascular endothelial growth factor (VEGF)-mediated angiogenesis and therefore, have a steroid-sparing (anti-edema), anti-angiogenic effect in glioblastoma patients[72]. If losartan does work as an anti-angiogenic agent, its use in unselected patients should mirror the results of other anti-angiogenic agents such as bevacizumab). Indeed, and similar to the results with losartan[63], bevacizumab did not prolong survival in unselected patients with glioblastomas as reported in the RTOG-0825 and AVAGlio trials[73]. To this end, it is unlikely that all patients in the treatment-arm will have a tumor that benefit from normalization of tissue to such an extent that it will reach statistical significance compared to the control group. This hypothesis is supported by a range of retrospective analyses reporting prolonged survival in glioblastoma patient subgroups with a favorable vascular signature for anti-angiogenic therapy[74-76].

The rationale for our study is based on our preliminary human data[77] that corroborate pre-clinical findings[50] of restored perfusion from ARBs and not from other anti-hypertensives that do not target the renin-angiotensin system. Here, anti-hypertensives where given to counteract high blood pressure caused by anti-angiogenic therapy[76]. To assess these effects, we will also create an RPI signature ('Restricted Perfusion Imaging'; distribution of image voxels) of a patient's tumor and its vascular restrictions in vivo by combining MRI-based measures of vascular efficiency from MR perfusion with those of tissue solid stress from MR Elastography (**Figure 1**). We hypothesize that in patients where the majority of the tumor vasculature is close to- or within the normal range at baseline and/or following ARBs will return improved radiographic and clinical responses to radio-chemotherapy compared to patients with a vascular profile suggesting mostly impaired vasculature. This is supported by our data showing that patients with glioblastoma or metastases to the brain with non-infiltrative, 'pushing' growth patterns[78] induce more vascular restrictions on brain tissue compared to tumors with an infiltrative phenotype[71,79,80]. Moreover, ARBs are believed to modulate the microvasculature by two alternative mechanisms; (**1**) either by vessel decompression from alleviated solid stress[50] or (**2**) they potentiate anti-angiogenic therapy directly through AngI inhibition[54]. Because brain tumors are particularly heterogenous[79,81], we hypothesize these mechanisms are not exclusive, but coexisting or even dynamic[82], and that RPI may reveal how ARBs modulate the microenvironment of human tumors (**Figure 2**).

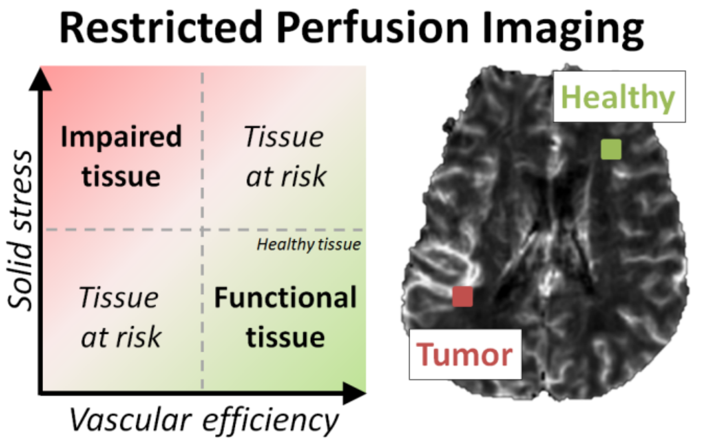

Figure 1: Conceptual schematic of the Restricted Perfusion Imaging (RPI) paradigm. Relative to the average levels of healthy-appearing tissue, we propose to map impaired tissue in cancer as a function of high solid stress from MR Elastography and low 'vascular efficiency' (=<u>vessel caliber x perfusion</u>) from perfusion MRI and VAI.

Moreover, recent clinical observations suggest that immunotherapy aggravate brain edema that in turn requires steroids[83-85]. Because steroids cause immuno-suppression[19,86], treating brain cancers with checkpoint blockers may be inherently limited and explain the conflicting clinical data[45,46,87]. Moreover, elevated solid stress also causes immuno-suppressive hypoxia[19,88]. Taken together, this constitutes a formidable barrier to therapy and an agent that can control edema without causing immuno-suppression is urgently needed. Because ARBs may alleviate solid stress, reduce hypoxia and decrease edema in selected brain tumors[72] - and thus potentially also reduce the need for steroids[67] - we hypothesize ARBs can convert an immuno-suppressive microenvironment into an immuno-stimulatory one. Imaging is the only way to assess brain edema and we will perform an open-label MRI-study on losartan utilizing the study pipeline from our ongoing imaging study ('TREATMENT'; Clinicaltrials.gov Identifier NCT03458455) in patients with brain metastases from lung cancer receiving chemotherapy or immunotherapeutic drugs pembrolizumab or atezolizumab[45]. With the potential limited benefit of losartan on edema in unselected patients with glioblastomas[63], instead, we hypothesize the higher frequency of observed vascular impairment in patients with brain metastases constitute a more favorable target for this intervention[71]. This approach is also supported by other independent data by our group showing distinct vascular signatures in brain metastases that indicate high treatment-potential for ARB therapy[89,90].

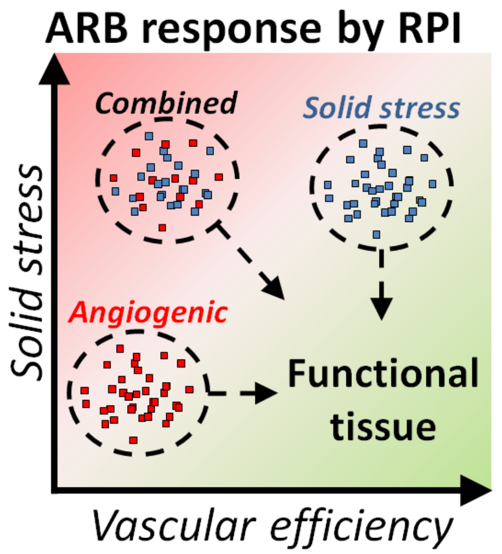

Figure 2: RPI reveals angiotensin II receptor blocker (ARB) response. RPI voxels dominated by a **left-right** shift following ARBs suggests an <u>angiogenic</u>-dominated mechanistic response. An **up-down** shift however, suggests a <u>solid stress</u>-dominated mechanistic response. We believe ARBs will induce both.

1.5 Treatment administration

Study administration of losartan will be inside the dose terms of the marketing authorization. Losartan (Cozaar®) is sold as 12.5mg, 25mg, 50mg or 100mg film-coated tablets containing the equivalent dose of losartan potassium. To assess any dose-response relationship between losartan and our imaging-based primary endpoint, we will perform an individual stepped-wedge, randomized, assessor-blinded, dose-finding treatment administration design on three dose levels. All inside the dose terms of the marketing authorization, this includes a ‘low dose’ (25mg), a ‘standard dose’ (50mg) and a ‘high dose’ (100mg). All dose regimes are reported safe and well tolerated[58]. To adhere to the treatment cycles of the standard-of-care cancer therapies and institutional imaging intervals., *losartan* (tablet) will be administered every day, with 2 weeks (14 consecutive days) defined as a treatment cycle for patients with glioblastoma (**Study A)** and 3 months (90 consecutive days) for lung cancer patients with metastases to the brain (**Study B**).

1.6 Treatment protocol

This is an open-label, single institutional phase II trial with an individual stepped-wedge, randomized, assessor-blinded, dose-finding design on three indications. The study design is described in detail in sections 5.2-5.3. The participant, the trial site personnel and the Sponsor are not blinded to treatment. Drug identity (name, strength) is included in the label text. The patient will be assigned to a study cohort by randomization.

The treatment protocol will also include a run-in period in patients with recurrent and newly diagnosed glioblastoma (**Study A** – both cohorts) to assess the **efficacy**, **safety**, **futility** and **feasibility** of the IMP in a representative study population. The run-in period and subsequent interim analysis will constitute an alternative to a traditional study pilot and allow us to;

- Estimate an accurate **effect size** of losartan on our primary endpoints
- Monitor **safety** by the AEs described in section 8.1, and as collected at every hospital visit.
- Estimate the potential **futility** of the IMP and study primary endpoint
- Perform a subjective evaluation by the investigators and sponsor on the **feasibility** of our treatment design and imaging plan.

Participants from the run-in cohort will be included in the final study analysis.

1.7 Risk-benefit of study participation

Besides intervention with the IMP/losartan, the patients included in the study will have some more MRI examinations and a more standardized follow-up than patients outside the study. For many patients this is a wanted side effect as many glioblastoma patients ask for a centralized follow-up. The primary endpoint will probably not affect outcome for included patients, whereas the addition of the IMP/losartan might lead to a better survival.

Patients with glioblastoma have a dismal prognosis and receive standard treatment including chemotherapy and radiotherapy. Both these treatment modalities carry a risk of serious treatment-induced disease and side effects, but as they are evidence-based and with the incurable nature of glioblastoma, their use is justified. The potential side effects of the IMP/losartan are listed in section 8.2 and the most common/important ones are hypotension and electrolyte disturbances. It is unlikely that these will affect quality of life for treated patients and the risk-benefit profile is far better than most other anti-neoplastic drugs. To reduce the risk of losartan-related side effects, patients will be monitored for these potential side effects with vital signs including blood pressure and pulse, as well as weekly blood tests. These procedures are part of routine treatment of patients with glioblastoma today.

2 STUDY OBJECTIVES AND RELATED ENDPOINTS

	Objectives	Endpoints	Assessments
Primary	To assess the dose-response relationship of the angiotensin II receptor inhibitor losartan on perfusion and mechanical solid stress in brain tumors	Primary efficacy endpoint: Change from baseline in the radiographic biomarker relative cerebral blood flow (rCBF) in the primary tumor area, and normalized to reference tissue [Time Frame:	Section 7.1

		Study A (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study B** (lung patients with metastases to the brain) Baseline, Day 90-14, Day 180-14, Day 270-14] Change from baseline in the radiographic biomarker tissue stiffness, as derived from the shear wave modulus G of the MR-elastography readout in the primary tumor area, and normalized to reference tissue [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44 **Study B** (lung patients with metastases to the brain) Baseline, Day 90-14, Day 180-14, Day 270-14]	
Secondary	To assess the dose-response relationship of losartan on experimental radiographic characteristics in patients with glioblastoma and metastases to the brain	Change from baseline in experimental radiographic biomarkers from MRI, including total volume of enhancing tumor, the total volume of tumor edema, perfusion MRI, diffusion MRI, MR Elastography, and normalized to reference tissue [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 78-8/+1, **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 148-8/+1, **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-14, Day 180-14, Day 270-14] Change from baseline in neurologic performance scores by Karnofsky Performance Score (KPS), Eastern Cooperative Oncology Group (ECOG) and/or Neurologic Assessment in Neuro-Oncology (NANO) scores. [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 42-44, Day 85±4, Day 169±4.	Section 7.1

	Study A (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 42-44, Day 71±4, Day 155±4, Day 239±4. **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7] Best overall response of radiographic status on MRIs of intracranial disease or corresponding treatment response by the Response Assessment in Neuro-Oncology (RANO) criteria (occurrence of radionecrosis, pseudoprogression or tumor progression). [Time Frame: **Study A** (patients with recurrent glioblastoma) Day 78-8/+1, Day 162-8/+1, **Study A** (patients with newly diagnosed glioblastoma) Day 148-8/+1, Day 232-8/+1,. **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-14, Day 180-14, Day 270-14, Day 360±7.]
To assess the dose-response relationship of losartan as add-on to standard treatment on clinical or patient reported endpoints in patients with glioblastoma and metastases to the brain	Total dose and change in steroids dosage during study treatment duration. [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 57±4, Day 85±4, Day 113±4, Day 127±4 for Lomustine and Day 141±4 for Temodal, Day 169±4. **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 28-30, Day 42-44, Day 71±4, Day 99±4, Day 127±4, Day 155±4, Day 183±4, Day 211±4, Day 239±4 **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7] Change from baseline in the European Organization for Research and Treatment of Cancer (EORTC) quality of life questionnaire (QLQ BN20) core 30 version 3.0. [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 42-44, 169±4. **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 42-44, Day 155±4, Day 239±4. **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7]

	Change from baseline in experimental biomarkers from blood and serum, including vascular endothelial growth factors (VEGF-A, VEGF-C, VEGFR2, CD31, FGF, PDGF, IGFBP7), angiopoietins (ANG-1, ANG-2, ANGPTL4), placental growth factors (PIGF), epidermal growth factor receptors (EGF, TGF-alpha), hyaluronan levels, collagens levels (collagen IV alpha 1, MMP-10, Tenascin C, CD163, CA9, THBS2), and inflammatory cytokines (cxcl9-11, CCL2, CX3CL1, ErbB2, Fas Ligand/TNFSF6, GITR/TNFRSF18, TNF-alpha, Granzyme B, IFN-gamma, IL-1beta, IL-2, IL-6, IL-8, IL-12). [Time Frame: **Study A** (patients with recurrent glioblastoma) Baseline, Day 14-16, Day 42-44, Day 90±4, Day 155±4. **Study A** (patients with newly diagnosed glioblastoma) Baseline, Day 14-16, Day 42-44, Day 225±4. **Study B** (lung cancer patients with metastases to the brain) Baseline, Day 90-7, Day 180-7, Day 270-7, Day 360±7] Six-months progression free survival (6M-PFS) [Time Frame: Day +180] Progression free survival (PFS) [Time Frame: up to 24 months] Overall survival (OS) at 12 and 24 months [Time Frame: at 12 months, 24 months] Overall survival [Time Frame: up to 24 months]
To assess the safety of losartan treatment in in patients with glioblastoma and metastases to the brain	Drug tolerance of IMP (NCI-CTCAE v4.0) according to study and follow-up period, including number and frequency of losartan treatment emerging adverse events (TEAE), vital signs (, blood pressure, pulse) and conventional laboratory blood parameters. [Time Frame: **Study A** (patients with recurrent glioblastoma) up until 168 + 14 days **Study A** (patients with newly diagnosed glioblastoma) up until 238 + 14 days **Study B** (lung cancer patients with metastases to the brain) up until 270 + 14 days]

Other	To further explore and develop magnetic resonance imaging methods and their correlation to clinical and blood/serum data with the objective to identify personalized treatment options in patients with glioblastoma and metastases to the brain	There will be no pre-specified endpoints for this objective. The objective will be explored using a wide variety of methods.

3 OVERALL STUDY DESIGN

The ImPRESS study is an open-label, blinded assessor, single center, multi-dose individual-randomized stepped-wedge trial on three indications of brain cancers.

Eligible patients will be randomized to different treatment regimens (dose and start time) of losartan according to indication (recurrent glioblastomas, newly diagnosed glioblastomas or metastases to the brain), on top of standard treatment as described in chapter 5.

Study Period	Estimated date of first patient enrolled: **01-JUN-2019** Anticipated recruitment period: **60 MONTHS** Estimated date of last patient-last visit completed: **01-MAY-2025**
Treatment Duration:	Expected treatment duration per patient: Patients with recurrent glioblastomas: 14 to 168 days (matching 6 cycles of chemotherapy). Patients with newly diagnosed glioblastomas: 14 to 238 days (matching concomitant radiation-chemotherapy and 6 cycles of chemotherapy). Patients with metastases to the brain: 90 to 270 days (matching 9 months of concomitant therapy)

3.1 Duration of follow-up

After the end of the final treatment cycle, each participant will be followed for 14 days for adverse event monitoring (AE will be collected for 14 days after the final treatment, regardless if the participant received IMP or not). Participants will be followed for PFS and OS for a period up to 24 Months. Participants who discontinue for reasons other than progressive disease will have post-treatment follow-up for disease status until disease progression, initiation of a non-study cancer treatment, withdrawal of consent or to they are lost to follow-up, whichever comes first. After documented disease progression each participant will be followed by the study nurse for overall survival until death, withdrawal of consent, or end of the study, whichever occurs first.

4 STUDY POPULATION

4.1 Selection of Study Population

This is public-sponsored, single-institution study at Oslo University Hospital, Oslo, Norway, and hosted by the Department of Diagnostic Physics, Division of Radiology and Nuclear Medicine. Serving 2.7 million people (~50% of Norway), Oslo

University Hospital personnel diagnoses, operates and performs life-long follow-up of >200 patients with glioblastoma (~120) and metastases to the brain (~100) each year.

Study A: Glioblastoma is one of the deadliest forms of cancer and the primary goal of treatment is simply to decelerate tumor growth[40,41]. Still, after decades of research, standard-of-care includes surgery, radiotherapy and chemotherapy and glioblastoma remains an incurable disease. The ***ImPRESS-losartan*** personnel have extensive imaging research experience with this patient group that allows for unmatched high-quality imaging data (i.e. minimal movement). Choosing brain cancer patients over other desmoplastic tumors types outside the brain is justified by the intriguing observation that glioma cell migration is selectively dependent on the stiffness of the surrounding tissue[91] and that the unique architecture of the healthy extra-cellular matrix in the brain allows for diffusion of larger molecules across the matrix[69]. The latter is especially intriguing because it makes the recovery-potential of the cancerous brain matrix high, and treated tissue will allow for high drug penetration of chemotherapy and other larger-molecule therapies. This is catalyzed by the compromised blood-brain barrier of leaky brain tumors, allowing for enhanced intra- to extra-vascular transport of blood-borne therapeutic molecules at the cancer site.

Study B: Brain metastases are increasingly common because many primary tumors eventually relapse and metastasize[40] and the aggressive nature of this cancer type makes metastases especially difficult to treat [37]. For sufficient patient enrollment, ***ImPRESS-losartan*** focus on the biggest cohort at Oslo University Hospital; patients with brain metastases from non-small-cell lung cancer (NSCLC). Most patients with NSCLC with brain metastases receive stereotactic radiosurgery (20-30Gy) to all or selected lesions as part of their treatment course. NSCLC patients with stage IV disease (all patients with brain metastasis have stage IV disease) receive systemic anti-neoplastic therapy, either combination treatment with immunotherapy and chemotherapy, immunotherapy only or chemotherapy only per clinical routine. The choice of systemic anti-neoplastic treatment for lung cancer patients in IMPRESS will be according to national guidelines.

4.2 Number of Patients

153 patients in total will be randomized in this trial, 54 patients with recurrent glioblastoma, 54 patients with newly diagnosed glioblastoma and 45 patients with metastases to the brain.

4.3 Inclusion Criteria

Patients will be recruited from the catchment area of the study site, Oslo University Hospital. All of the following conditions must apply to the prospective patient at screening prior to receiving study agent:

1. A histologically confirmed intracranial glioblastoma, WHO grade 4 (**Study A**), or a minimum of one radiographically confirmed metastasis to the brain from a primary non-small-cell lung cancer (**Study B**)
2. Ability to undergo an MRI exam, including administration of a standard clinical dose of an MRI-specific contrast agent of gadolinium or similar
3. Measurable intracranial disease (**Study A** – recurrent glioblastoma, and **Study B** only), defined as at least one lesion that can be accurately measured in at least one dimension as ≥10 mm with MRI – or – compromise more than 30 image voxels on perfusion MRI to ensure adequate parametric statistical assessments. For a perfusion MRI resolution of 1.2x1.2x5mm, this equals a tumor volume of 0.2 cubic centimeters (cc).
4. Age ≥18 years
5. Eligible for administration of the active substance (losartan) in concordance with study protocol, the criteria of the product label (Cozaar) and deemed fit for trial by the treating physician.
6. An ECOG performance status of ≤2 or equivalent KPS of ≥60%
7. Life expectancy from start of treatment of more than 3 months
8. Previous history of a neurosurgical procedure, at time of study inclusion (**Study A** only)

9. Scheduled for chemotherapy and/or radiotherapy and/or stereotactic radiosurgery (**Study A** – recurrent glioblastoma), neurosurgery, radiotherapy and chemotherapy (**Study A** – newly diagnosed glioblastoma), immunotherapy and/or chemotherapy and/or stereotactic radiosurgery (**Study B**)
10. Pre-study documentation on O^6-methylguanin-DNA-methyltransferase (MGMT) promoter methylation status and on the isocitrate dehydrogenase (IDH) gene mutation status of their disease (**Study A** only)
11. Organ functions of sufficient quality and robustness to undergo study treatment as determined by the study PI or designee
12. Female patients of childbearing potential (postmenarcheal, not postmenopausal (>12 continuous months of amenorrhea with no identified cause other than menopause), and no surgical sterilization) should use highly effective contraception and take active measures to avoid pregnancy while undergoing IMP treatment and for at least 14 days after the last dose. Birth control methods considered to be highly effective include combined (estrogen and progestogen containing) hormonal contraception associated with inhibition of ovulation (oral, intravaginal or transdermal), progestogen-only hormonal contraception associated with inhibition of ovulation (oral, injectable or implantable), intrauterine device, intrauterine hormone-releasing system, bilateral tubal occlusion, vasectomized partner, sexual abstinence when it is the preferred and usual lifestyle of the subject.
13. Ability to understand and the willingness to sign a written informed consent document

4.4 Exclusion Criteria

Patients will be excluded from the study if they meet any of the following criteria, including any contraindication listed on the Summary of Product Characteristics (SmPC):

1. Hypersensitivity to the active substance (losartan) or to any of the excipients
2. Patients on antihypertensive agents that cannot be substituted with losartan.
3. Patients on medication that may induce hypotension and/or increase potassium levels and/or cause metabolism-related pharmacokinetic drug-drug interactions with losartan. If medication with such effects can safely be discontinued or replaced, the patient can be included in the study after a washout period before baseline.
4. Patients with hepatic or renal impairment of any reason
5. Patients with symptomatic hypotension of any reason
6. Patients with primary hyperaldosteronism
7. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucosegalactose malabsorption should not take this medicine
8. Inadequate recovery from toxicity and/or complications of previous therapy as determined by the treating physician
9. Patients with evidence of recurrence inside the radiotherapy target volume less than 3 months since last radiotherapy fraction (**Study A –** recurrent glioblastoma only)
10. For **Study B** subjects only: A diagnosis of immunodeficiency or hypersensitivity to PD-1 inhibitors or any excipients.
11. Uncontrolled intercurrent illness including, but not limited to, ongoing or active infection, symptomatic congestive heart failure, coronary heart disease and cerebrovascular disease, unstable angina pectoris, cardiac arrhythmia, angioedema, intravascular volume depletion, or psychiatric illness/social situations that would limit compliance with study requirements, as determined by the treating physician
12. Patients suffering from aortic or mitral stenosis, or obstructive hypertrophic cardiomyopathy
13. Patients with known psychiatric or substance abuse disorders that would interfere with cooperation with the requirements of the study as determined by the treating physician
14. Pregnant or breastfeeding patient
15. Known additional active non-study related malignancy
16. Applies to **Study B** patients qualifying for immunotherapy only: Active autoimmune disease that has required systemic treatment in the last 2 years (including use of non-study related disease modifying agents, corticosteroids or

immunosuppressive drugs). Replacement therapy (e.g., thyroxine, insulin, or physiologic corticosteroid replacement therapy for adrenal or pituitary insufficiency, etc.) is allowed

17. Known history of Human Immunodeficiency Virus (HIV) (HIV 1/2 antibodies)

18. Unable to undergo brain MRI according to study protocol

5 STUDY TREATMENT

For this study losartan (Cozaar®), an angiotensin-II inhibitor, is defined as Investigational Medicinal Product (IMP). This is an open-label study and no active comparator or placebo will be used. The study design is an individual stepped-wedge, randomized, assessor-blinded, multi-dose, single-center trial on three indications.

5.1 Drug Identity, Supply and Storage

Drug identity

The name of the study medicinal product is Cozaar® 12.5mg, 25mg, 50mg or 100mg film-coated tablets containing the equivalent dose of losartan potassium. Losartan is a synthetic oral angiotensin-II receptor (type AT1) antagonist. The pharmacotherapeutic group is Angiotensin II antagonists, plain, ATC code: C09CA01.

Pharmaceutical form

• COZAAR 12.5 mg tablet: Blue, oval film-coated tablets marked 11 on one side and plain on the other.
• COZAAR 25 mg tablet: White, oval unscored film-coated tablets marked 951 on one side and plain on the other.
• COZAAR 50 mg tablet: White, oval film-coated tablets marked 952 on one side and scored on the other. The score line is not intended for breaking the tablet.
• COZAAR 100 mg tablet: White, teardrop-shaped film-coated tablets marked 960 on one side and plain on the other.
** Please refer to the storage and preparation instructions on the package leaflet.*

Clinical supplies disclosure

This trial is open-label; therefore, the participant, the trial site personnel, the sponsor and/or designee are not blinded to treatment. Drug identity (name, strength) is included in the label text; random code/disclosure envelopes or lists are not provided.

Storage, handling and preparation requirements

Losartan is a prescription drug and will be stored with the patient per label instructions.

Ordering

The study PI or designee will order losartan directly from the sponsor's pharmacy.

Accountability

The PI or designee are responsible for keeping accurate records of the clinical supplies received from the pharmacy, the amount dispensed to and returned by the participants and the amount remaining at the conclusion of the trial.

Destruction and return

Upon completion or termination of the study, and after independent monitoring approval of accountability, all unused and/or partially used investigational product will be destroyed at the site per institutional policy.

5.2 Dosage and Drug Administration

For patients on study treatment, *losartan* (tablet) will be administered every day, with 2 weeks (14 consecutive days) defined as a treatment cycle for patients with glioblastoma and 3 months (90 days) defined as a treatment cycle for lung cancer patients with metastases to the brain. The reason for this difference in treatment regimen is to adhere to the treatment cycles of the conventional cancer therapies and available institutional imaging intervals. We hypothesize that losartan improves the effect of traditional cancer treatment, while administration of losartan alone has little effect on cancer patients. For glioblastomas, this corresponds to a treatment cycle of 6 weeks + 2 days (44 days) (2 days are added for logistical reasons). For brain metastases on either chemotherapy or combined chemo- and immunotherapy, this corresponds to 9 months (270 days).

According to the manufacturer's SmPC (Cozaar), the active, mechanistic and pharmacological response to losartan is expected to be quick (within hours / days). For reference, with daily administration, maximum anti-hypertensive effect of losartan is achieved 3-6 weeks after onset. This corresponds to a treatment period of 6 weeks for glioblastoma, with the possibility of capturing a potential escalation phase for the effect of losartan at 2 weeks and a plateau with maximum effect at 4-6 weeks. For brain metastases, this must be weighed against the treatment length of chemo and immunotherapy whose effects are expected over a period of 9 months. Patients with brain metastasis are admitted for consultation and radiographic examination (primary endpoint MR) every third month.

According to the manufacturer's SmPC (Cozaar), patients on daily administration of losartan should observe an increasing or sustained effect of the drug. When treatment with Losartan is stopped, a similar rapid end to the mechanistic and pharmacological response is expected, corresponding to a half-life of 2 hours and a terminal half-life (washout time) of 6-9 hours. However, long-term clinical effects of losartan on cancer patients cannot be ruled out, and in concordance with our secondary endpoints.

The stepped-wedge design is proposed to assess any dose-response relationship of the IMP. To assess any additional long-term beneficial effects of the IMP while receiving conventional therapy, patients in Study A will be randomized at day 42 to either 1) continued treatment until day 168 (Study AR) or day 238 (Study AN) with the current dose or 2) losartan treatment discontinuation. For participants on 50mg and 100mg of the IMP, a dose reduction in steps of 25mg is allowed at the discretion of the study PI. No dose reduction is allowed for participants on 25mg of the IMP. In the case of a dose reduction, the patient is allowed to stay on trial, but is excluded from analysis of the primary endpoint.

Treatment will be administered on an outpatient basis. No investigational or commercial agents or therapies other than those described in this document may be administered as part of the study with the intent to treat the participant's malignancy. Study medicine should be administered at the same time every day, and with or without food at the participants' discretion or per label instructions. The tablet should be swallowed in one piece with a glass of water or similar.

The losartan treatment to be used in this study is as follows;

Indication	Dose/Potency	Dose Frequency	Route of Administration
Patients with recurrent glioblastoma	25mg, 50mg or 100 mg by randomization	Q1D	Oral
Patients with newly diagnosed glioblastoma	25mg, 50mg or 100 mg by randomization	Q1D	Oral
Lung cancer patients with metastases to the brain	50 mg	Q1D	Oral

The study treatment will be administered to the participant by authorized site personnel only.

Patients on losartan

Patients diagnosed with glioblastoma and on losartan for any reason from beforehand can be included in the study. These patients will be subject to a wash-out period of 3 days before study initiation.

5.3 Duration of Therapy

- **Study A (recurrent glioblastoma – short form: 'AR')**

A maximum of twelve cycles of IMP (losartan) will be given. The stepped-wedge design is proposed to compare participants on- and off- study IMP and assess any dose-response relationship of the IMP. To assess any long-term beneficial effects of the IMP, some patients will receive the IMP for a longer period than only 16–44 days. Patients in groups AR 3_1, AR3_2 and AR 3-3 will all receive extended IMP treatment, whereas patients in groups AR1_1, AR1_2, AR1_3, AR2_4, AR2_5 and AR2_6 will be randomized to short or extended treatment with the IMP. This leads to a distribution with 1/3 of patients with no extended IMP treatment, 1/3 extended lower doses of IMP (1/6 25 mg and 1/6 50 mg), and 1/3 extended high dose of IMP.

Group	IMP dose	Baseline	Days 1 to 14	Days 15 to 28	Days 29 to 44	Days 45 to 168	Days > 168
AR1_1	25 mg						
AR1_2	25 mg						
AR1_3	25 mg						
AR2_4	50 mg						
AR2_5	50 mg						
AR2_6	50 mg						
AR3_1	100 mg						
AR3_2	100 mg						
AR3_3	100 mg						

Note. Gray is no treatment, green is treatment with losartan, green wedge is randomized to either losartan or no treatment. IMP; Investigational Medicinal Product.

- **Study A (newly diagnosed glioblastoma – denoted 'AN')**

A maximum of seventeen cycles of IMP (losartan) will be given. The stepped-wedge design is proposed to compare participants on- and off- study IMP and assess any dose-response relationship of the IMP. To assess any long-term beneficial effects of the IMP, some patients will receive the IMP for a longer period than only 16-44 days. Patients in groups AN3_1, AN3_2 and AN 3-3 will all receive extended IMP treatment, whereas patients in groups AN1_1, AN1_2, AN1_3, AN2_4, AN2_5 and AN2_6 will be randomized to short or extended treatment with the IMP. This leads to a distribution with 1/3 of patients with no extended IMP treatment, 1/3 extended lower doses of IMP (1/6 25 mg and 1/6 50 mg), and 1/3 extended high dose of IMP.

Group	IMP dose	Baseline	Days 1 to 14	Days 15 to 28	Days 29 to 44	Days 45 to 238	Days > 238
AN1_1	25 mg						
AN1_2	25 mg						
AN1_3	25 mg						
AN2_4	50 mg						
AN2_5	50 mg						
AN2_6	50 mg						
AN3_1	100 mg						
AN3_2	100 mg						
AN3_3	100 mg						

Note. Gray is no treatment, green is treatment with losartan, green wedge is randomized to either losartan or no treatment. IMP; Investigational Medicinal Product.

- **Study B (brain metastases – denoted 'BM')**

A maximum of three cycles will be given. The protocol will use a standard (50mg/daily) dose of losartan only[58]. The accrual rate at Oslo University Hospital and the long follow-up period for patients on immunotherapy suggest a dose-finding scheme similar to that of Study A will be outside the scope of the current project period. The stepped-wedge design is proposed to compare participants on- and off- study IMP.

Group	IMP dose	Baseline	Days 1 to 90	Days 91 to 180	Days 181 to 270	Days > 271
BM1	50 mg					
BM2	50 mg					
BM3	50 mg					

Note. Gray is no treatment, green is treatment with losartan. IMP; Investigational Medicinal Product.

5.4 Premedication and Monitoring

There will be no need for premedication. Blood pressure and pulse are monitored at all hospital visits in combination with registration of any adverse events. Please refer to flow charts in section 6.3.

5.5 Schedule Modifications

Depending on patients' experience and the severity and content of any reports of adverse events, losartan may be interrupted, discontinued or dose adjusted at any time at the sole discretion of the treating physician.

5.6 Non-study specific treatment regimens

The non-specific treatment plan, including the time and type of treatment, as well as the choice of concomitant, adjuvant or salvage therapy, is solely within the responsibility of the treating physician. No patient should be suggested for inclusion in the study for any other reason than what is deemed the optimal treatment regime for that patient. All non-specific study agent doses and duration of treatments are according to manufacturer prescription labels and recommended standard-of-care. No special precautions or modifications of the standard treatment plan will be performed based upon inclusion in the study.

Surgery or stereotactic radiosurgery (SRS)

For inclusion in the primary endpoint analysis, surgery or SRS is not allowed during the study IMP cycles (Day 1 - 270), this applies also to participants randomized to start IMP treatment at Days 91 or 181. Any need of repeated surgery or SRS of study participants during the study follow-up period is at the sole responsibility of the treating physician. Indications for repeated surgery or SRS include new lesions, increasing neurological deficits, raised intracranial pressure, radiographic tumor progression, and epileptic seizures[41]. These events will be recorded as disease progression. Participants receiving surgery or SRS during study treatment will stop IMP administration. All surgical or SRS events will we reported as part of the patients' clinical study information. Patients receiving additional surgery or SRS may still be included in the analysis for secondary endpoints and monitored for follow-up up until 24 months.

Stupp regimen (Study A – newly diagnosed glioblastomas)

The Stupp regimen is the standard of care for the treatment of patients with newly diagnosed glioblastomas after surgery[92]. It consists of fractionated radiotherapy and concomitant chemotherapy with temozolomide, an alkylating agent, and followed by six additional courses of adjuvant temozolomide:

- Radiotherapy: a total of 60 Gy in fractions of 2 Gy per day (Monday to Friday) over 6 weeks
- Temozolomide (during radiotherapy): 75 mg/m^2 body surface area (BSA) per day for 7 days per week
- Six adjuvant temozolomide cycles (following radiotherapy): 150-200 mg/m^2 BSA per day for 5 days, in the course of

28 days (2 study cycles).

The radiotherapy planning softwares (Masterplan and Raystation) at the trial site allows us to compare the dose distribution maps with MRI[93] on a pixel-by-pixel level in patients with newly diagnosed glioblastomas. The dose distribution maps are created and managed independently of the ***ImPRESS-losartan*** study and we will retrospectively collect the dose distribution maps from the hospital's radiotherapy planning software for comparison with the study MRI exams.

Chemotherapy (Study A – recurrent glioblastomas)

For patients with recurrent glioblastomas, temozolomide is administered as monotherapy in six cycles consisting of 150-200 mg per square meter for 5 days during each 28-day cycle. Alternatively, for patients with glioblastomas progressing during primary treatment or recurring within three months off primary treatment, lomustine (CCNU) is administered as anti-neoplastic monotherapy in four cycles (two plus two) consisting of 110 mg per square meter BSA as a one-day dosage during each 42-day cycle. Maximum dose for lomustine is 200 mg, but the dose may be adjusted in case of hematological toxicity, either previously or during treatment.

Systemic anti-neoplastic therapy (Study B)

Lung cancer patients included will receive systemic treatment which consists of chemotherapy, immunotherapy or a combination of those two.. Treatment is based on histology, PD-L1 expression, mutations and if the brain metastases are present at primary diagnosis or develop at a later stage.

The choice of chemo- and immunotherapeutic agents used in ImPRESS will be according to national guidelines and overall clinical evaluation of the treating physician.

The chemotherapeutic standard of care for patients with NSCLC is carboplatin in combination with vinorelbin or pemetrexed or equivalent analogs, and administered per label instructions.

Some patients will receive a combination of chemotherapy and immunotherapy. Different regimens are used, but most commonly used is a combination of carboplatin (AUC 5), pemetrexed or paclitaxel and pembrolizumab. This might, however, vary and might change according to changes in national guidelines.

5.7 Concomitant Medication and Supportive Care Guidelines

All treatment that the treating physician considers necessary for a participant's welfare may be administered at the discretion of the PI in keeping with the community standards of medical care. All concomitant medication will be recorded in the electronic CRF by the responsible research nurse. If changes occur during the study period, documentation of drug dosage, frequency, route, and date may also be included in the central registration log. Participants should maintain a normal diet unless suggested otherwise by the treating physician.

Medications or treatments specifically prohibited in the exclusion criteria are not allowed during the study. If there is a clinical indication for one of these or other medications or treatments specifically prohibited during the trial, discontinuation of the study treatment (losartan) may be required. The final decision on any study treatment cessation rests with the study PI and/or the participant's primary care physician.

Participants are prohibited from receiving the following therapies during the Screening and Treatment phases (including retreatment for post-complete response relapse) of this study:

- ARBs, ACE inhibitors or any other anti-hypertensive medicine or analog not specified in this protocol
- Losartan administered with other antihypertensive agents, including with diuretics
- Tyrosine kinase inhibitors, vascular endothelial growth factor (VEGF) inhibitors or analogous vascular anti-cancer agents

All concomitant medication (including vitamins, herbal preparation and other “over-the-counter” drugs) used by the patient will be recorded in the patient’s file, but not the eCRF.

5.8 Compliance and Drug Accountability

A participant who forgets or for some other reason does not take the IMP at any single on-treatment day is allowed to continue on trial as long as the dose regimen is continued. Participants will be instructed not to compensate for the missing administration, but continue to take the IMP at the same time every day. All participants will receive a personal diary card to record their self-administration of the IMP. The participants will be asked to bring the diary card and the IMP boxes, both empty- and not-empty boxes, to every study visit. The study nurse will monitor the diary card and perform a drug accountability. At the end of the study compliance of IMP intake will be calculated.

All participants will receive a personal diary leaflet to record their self-administration of the IMP. The diary will be available in Norwegian language and include information as follows;

- The IMP is to be administered in the designated tablet dose and at the same time every day
- The IMP is to be administered with or without food at the participants’ discretion
- The IMP tablet should be swallowed in one piece with a glass of water or similar
- All administrations of IMP must be recorded in the diary including date and time. Also, any AE’s or other relevant experiences are to be recorded for that event.
- If the IMP is not taken on any single day, the participant must continue of study protocol on the following day
- No additional administration of the IMP is allowed on the second day to compensate for the missing administration of the IMP on the previous day
- In the case of a breach of study protocol, or any unaccepted or intolerable AE’s, the participant must contact the study PI (contact info available) and/or seek professional assistance at the nearest hospital or emergency center.

5.9 Drug Labeling

The investigational product will have a label permanently affixed to the outside and will be labeled in accordance with ICH GCP and national regulations, stating that the material is for the clinical trial **ImPRESS-losartan** / investigational use only. The study-specific label will not cover the following information per manufacturer label:

- Trade name and active substance (losartan)
- Dosage
- Number of tablets
- Temperature for correct storage
- Batch number and expiration date
- Manufacturer by name and address
- The medicine should be kept out of reach of children

The study-specific label will also include blank lines (in Norwegian language) for:

- Patient’s name
- Name of prescribing doctor
- Dosage
- Date dispensed

- Protocol code
- Patient's subject number
- Telephone number to the study PI

5.10 Participant Numbering

Potential participants for this study will receive a screening number. After the participant has been found eligible and signed the Informed Consent Form, the screening number will be replaced by a unique subject number. Screening number and subject number can only be used once and not reused for any other participant.

6 STUDY PROCEDURES

6.1 Molecular pre-screening

Regardless of this study and collected as part of the clinical routine at the trial site, potential eligible patients of study A will have documentation on the O^6-methylguanin-DNA-methyltransferase (MGMT) promoter methylation status and on isocitrate dehydrogenase (IDH) gene mutation status of their disease[94-97].

6.2 The Screening Visit

The screening visit is performed by the study PI or designee before study-treatment onset. The study-approved informed consent form must be signed and dated before any screening procedures are performed. Procedures which are part of the clinical routine during the initial diagnostic work-up of the patient may be performed before obtaining study consent. A copy of the informed consent form must be given to the patient or to the person signing the form. The study PI or designee must record the date when the study informed consent was signed in the patient file. Patients will be evaluated against study inclusion (section 4.3) and exclusion criteria (section 4.4), as well as any other safety assessments deemed relevant by the study PI or designee. The consent procedure is described in section 6.6.

If screening assessments occur within 2 weeks before start of study-treatment onset Cycle 1 Day 1, then they may serve as the baseline and study test and do not need to be repeated. This will be considered the baseline clinical evaluation. Demographic information and baseline clinical characteristics as described in section 6.6 will be collected at the screening visit.

6.3 Registration procedures

Per referral by the treating physician, or following radiographic examination by the designated neuroradiologist prior to the baseline MRI, the study PI or designee may suggest eligibility of potential patients according to the protocol-specific eligibility checklist

All potential candidates for study inclusion are entitled to a minimum of 24 hours for informed consent. Potential candidates are free to sign the informed consent form at any time following oral and written information about the study.

Registration process at Oslo University Hospital

The responsible research nurse is accessible on Monday through Friday, from 8:00 AM to 4:00 PM (GMT+1). Patients may be registered after hours at the discretion of the study PI or designee.

- To be eligible for registration to the protocol, the participant must meet all relevant (**Study A** versus **Study B**) inclusion and exclusion criterion as determined during the study screening visit.

- Eligible patients will be asked to participate in the study by the PI or designee.
- Obtain written informed consent from the participant prior to the performance of any protocol specific procedures or assessments. Informed consent will be obtained after the study has been fully explained to the patient and before the conduct of any study specific imaging procedures or assessments.
- The patients' inclusion date in the study will be noted in the study electronic record (eCRF)
- The patients' date of consent and inclusion in the study will be noted in the electronic medical record

6.4 Information to be collected on Screening Failures

A patient who signed an informed consent form and has started with study specific procedures but is not randomized for any reason will be considered a screening failure. The demographic information, informed consent, and the reason for screening failure must be recorded in the eCRF by using the screening number

6.5 Flow Charts of Study Protocol

Please see **APPENDIX A** (**Study A** – recurrent glioblastoma), **APPENDIX B** (**Study A** – newly diagnosed glioblastoma), and **APPENDIX C** (**Study B** – brain metastases).

6.6 By Visit

6.6.1 Before Randomization

At first visit and/or when relevant, the information collected will include the patients' personal identification number, general patient demographics, clinical status with relevant medical history and current medical conditions, relevant concomitant medications, diagnosis and extent of tumor, baseline tumor mutation status (MGMT and IDH for Study A), and details of prior anti-neoplastic treatments. In addition, all females of childbearing potential will have a serum pregnancy test during screening and before randomization. If positive, the patient is not eligible for the study.

Clinical status and relevant medical history

The patients' clinical status is determined by assessing the following clinical information:

- Weight with date of the exam
- Clinical measures on neurologic status (KPS, ECOG, NANO) with date of the exam
- Clinical information on disease progression with date (**Study A** – recurrent glioblastoma)
- Medical history concerning or relevant to the treatment of the brain tumor
- Any non-research MR exams related to the diagnosis of the brain tumor (section 6.6.2)
- Blood pressure tests (section 6.6.2)
- Blood test relevant for the treatment of the brain tumor (section 6.6.2)

Physical examination

A complete physical examination will be conducted by the study PI or designee at screening/baseline and will include examination of skin, lungs, heart, abdomen, lymph nodes, extremities, and neurological status. If indicated based on medical history and/or symptoms, further clinical tests will be performed. Significant findings that were present prior to the signing of informed consent must be included in the patient file and the eCRF. Height is measured in centimeters (cm) and body weight in kilogram [kg] (to the nearest 0.1 kg) in indoor clothing, but without shoes. Height information will be collected at the screening visit only.

Neurologic performance status

Assessment of KPS and ECOG performance status (**APPENDIX D**) and/or the Neurologic Assessment in Neuro-Oncology (NANO) scale[98] (**APPENDIX E**) will be performed at screening.

Concomitant medication
All treatment that the study PI or designee considers necessary for a participants' welfare may be administered at the discretion of the investigators in keeping with the community standards of medical care. All concomitant medication will be recorded in the eCRF by the responsible research nurse. If changes occur during the study period, documentation of drug dosage, frequency, route, and date must also be included in the eCRF.

Radiographic tumor evaluation
Participants are evaluated for intracranial disease before treatment onset by the study PI and with the aid the study neuroradiologist at will. Patients may have measurable (radiographic) and non-measurable disease (clinical). Traditional radiographic response and progression will be evaluated in this study using the modified Response Assessment in Neuro-Oncology (RANO) criteria[99,100]. A trained neuroradiologist blinded to clinical and study data will outline the contrast agent enhancing lesion on the MR images as well as the surrounding area of FLAIR hyperintensity for assessment of intracranial edema. Lesion volumes are estimated using a volumetric approach[101] that summarize pathological image voxels. The radiographic tumor evaluation will be recorded in the eCRF by the study PI or the responsible research nurse.

- **Measurable disease** is defined as what that can be accurately measured in at least one dimension (longest diameter to be recorded) as ≥10 mm with MRI – or – affect more than 30 image voxels on perfusion MRI to ensure adequate parametric statistical assessments. For the resolution of a perfusion MRI measurement, 1.2x1.2x5mm, this equals a tumor volume of 0.2cubic centimeters (cc). All RANO-based tumor measurements must be recorded in <u>millimeters</u> in 2D (or decimal fractions of centimeters) and in cc in 3D.

For multifocal intracranial disease, no more than 3 measurable target lesions should be selected for evaluation (>=5mm in diameter in at least one dimension or a total tumor mass of >30 MRI voxels). Target lesions should be selected on the basis of their size (starting with the lesion with the longest diameter or biggest volume), be representative of other lesions and lend themselves to reproducible repeated measurements.

Non-measurable disease are unidimensional masses with margins not clearly defined, lesions with maximal diameter < 5mm, or a total tumor mass of < 30 MRI voxels. Participants who have evaluable lesions at baseline, but on follow-up MR examinations do not meet the definitions of measurable disease will be followed for non-measurable disease. The response assessment is based on the presence, absence, or unequivocal progression of the lesions. In the case of non-measurable disease, MRI-based values (radiographic) in regions outside and surrounding the focal point of the lesion may still be assessed.

Additional optional PET-CT imaging
Subjects included in the Study A cohort – recurrent glioblastoma, will be offered the possibility to simultaneously participate in another clinical imaging study, the ImPRESS – FORCE protocol. There will be a separate patient information leaflet and Informed Consent Form for this study. By signing the Informed Consent Form, the subject commits to two additional PET-CT exams performed at time point of screening and at day 15-17. The ImPRESS - FORCE protocol and patient information leaflet have been approved by the Ethics Committee: Norwegian REC approval number 2017/1875. The PET-CT imaging data collected will not be part of the efficacy assessment of this study and solemnly processed as part of the ImPRESS – FORCE protocol.

Exploratory blood and serum collection
We will collect blood tests for biobanking at strategic time points prior to and during treatment (**APPENDIX A-C**) for the purpose described in section 7.4. This will include one vial of serum and one vial of plasma as follows;

- **Serum:** The sample is centrifuged at 2450g for 15 minutes at room temperature within 1 hour after sampling. Each sample is split into four tubes of 1mL. Any remaining serum is filled in an extra tube. Stored at -80° Celsius.
- **Plasma:** The sample is centrifuged 1000g for 10 minutes at room temperature within 1 hour after sampling. Each sample is split into four tubes of 200-250µL, and excluding the last 0.5cm of the plasma sample to avoid buffy coat contamination. Any remaining plasma is filled into an extra tube. Stored at -80 ° Celsius. The buffy coat is labeled and stored separately in a dedicated cryotube at -80 ° C.

6.6.2 During Treatment

Physical examination

A short physical exam (examination of general appearance) performed by the study PI or designee will occur at scheduled study visits (section 6.5) unless the study PI considers a complete physical examination necessary. Significant new findings that begin or worsen after treatment onset will be recorded as AE in the patient's CRF. Weight information will be collected at every physical exam.

Neurologic performance status

Assessment of KPS and ECOG performance status (APPENDIX-D) and the NANO scale[98] (APPENDIX-E) will be performed by the study PI or designee at scheduled study visits (section 6.5) unless the PI considers a new assessment unnecessary.

Quality of life

Quality of life will be assessed by the study PI or designee using the EORTC QLQ BN20 'and QLQ-C30 forms at scheduled study visits (section 6.5).

MRI exam

All exams are performed on dedicated 3 Tesla Philips, Siemens and GE Healthcare MRI clinical scanners at the trial site. The ***ImPRESS-losartan*** imaging protocol is as follows:

- Rapid localizer scan for automatic slice positioning and improved longitudinal registration ('AutoAlign')[102]
- High-resolution, 3D anisotropic T2-weighted (fluid-attenuated inversion recovery - FLAIR)
- High-resolution, 3D anisotropic T1-weighted imaging with- and without gadolinium contrast (0.1mMol/kg)
- 2D diffusion tensor MRI to monitor cerebral water diffusion and the degree of fractional anisotropy of the diffusion process[103]
- 2D contrast-enhanced perfusion MRI (gradient-/spin-echo dynamic susceptibility contrast) that allows for VAI and vessel size imaging
- 2D spin-echo motion-sensitive MRE including an MR-compatible external wave generator and triggered by the scanner so that the image acquisition and the shear wave (within the frequency range; 10–100 Hz) are in sync[80].

Please refer to chapter 7 for details on image analyses. The expected time in the MRI scanner per exam is approximately one hour, which is within the clinically accepted time for this patient group. Per clinical routine at the trial site, at this stage all eligible patients are cleared for contrast-enhanced MRI following previous successful experiences with contrast-enhanced MRI exams without any allergic reactions or symptoms. Moreover, all eligible patients are scheduled to receive contrast-agent injections regardless of the ***ImPRESS-losartan*** imaging protocol.

The study imaging intervals and acceptance ranges are described in section 6.5.

In addition, at the treating physician's discretion, additional non-study specific MR exams may be acquired, and henceforth included within the ***ImPRESS-losartan*** study period for the purpose of determining transient and long-term radiographic responses, respectively. This additional data collection will continue until the patient is deemed unfit for continued imaging by MRI, withdrawal of consent, or the end of study, whichever occurs first.

Radiographic evaluation

Please refer to the descriptions of the experimental (section 7.1) and traditional (section 7.2) radiographic assessments of intracranial neoplastic disease

Vital signs and blood pressure tests

Vital signs include blood pressure and pulse measurement and will occur at scheduled study visits either at the trial site or at a local hospital (section 6.5) unless the study PI considers an assessment unnecessary. After the patient has been sitting for five minutes, with back support and both feet placed on the floor, systolic and diastolic blood pressure will be measured three times using an automated validated device, e.g. OMRON, with an appropriately sized cuff. The repeat sitting measurements will be made at minute intervals and the mean of the three measurements will be used. In case the cuff sizes available are not large enough for the patient's arm circumference, a sphygmomanometer with an appropriately sized cuff may be used.

Laboratory blood tests

Routine biochemistry, sample collection, handling etc. will be collected by the study nurse or designee either at the trial site or at a local hospital and processed in accordance with the hospital´s laboratory standard procedures. The different study groups included in this protocol will follow standardized blood sample protocols according to clinical practice for the participant's specific disease. Blood samples taken at the trial site will be collected for routine analysis, as part of the secondary endpoints, as well as for biobanking for future exploratory research as indicated in the informed consent form (section 7.4).

The following blood sample protocols will apply for patients with glioblastoma:

Study A **CNS New Male**	**Study A** **CNS New Female**	**Study A** **CNS Follow-up ("KONT")**
White blood cell (WBC) count	White blood cell (WBC) count	White blood cell (WBC) count
Hemoglobin	Hemoglobin	Hemoglobin
Leukocytes	Leukocytes	Leukocytes
Thrombocytes	Thrombocytes	Thrombocytes
ACTH	ACTH	ALT
ALT	ALT	Albumin
Albumin	Albumin	ALP
Alkaline phosphatase (ALP)	ALP	AST
AST	AST	Bilirubin
Bilirubin	Bilirubin	CRP
CRP	CRP	Glucose
FSH	FSH	GT
FT4	FT4	Potassium
Glucose	Glucose	Calcium
GT	GT	Urea
Potassium	Potassium	Chloride
Calcium	Calcium	Creatinine
Urea	Urea	LD
Chloride	Chloride	Magnesium
Cortisol	Cortisol	Sodium
Creatinine	Creatinine	
LD	LD	
LH	LH	
Magnesium	Magnesium	
Sodium	Sodium	
Prolactin	Prolactin	
Sex hormone binding globulin (SHBG)	TSH	
Testosterone	Estradiol	
TSH		

The following blood sample protocols will apply for patients with brain metastases:

Study B **LUNG New**	**Study B** **LUNG Follow-up ("KONT")**	**Study B** **LUNG Follow-up ("IMMUN")**
Hemoglobin	Hemoglobin	ALanine Amino-Transferase (ALAT)
Erythrocyte Volume Fraction (Hct)	Erythrocyte Volume Fraction (Hct)	Albumin
Platelets	Platelets	Alkaline phosphatase (ALP)
Differential counting	Differential counting	Amylase, pancreas
Leukocytes	Leukocytes	ASparine Amino-Transferase (ASAT)
Sodium	Sodium	Basophilic granulocytes
Potassium	Potassium	Bilirubin
Calcium	Calcium	Calcium
Bilirubin total	AST	C-reactive protein
AST	ALT	Creatinine

Study B LUNG New	Study B LUNG Follow-up ("KONT")	Study B LUNG Follow-up ("IMMUN")
ALT	LD	Eosinophilic granulocytes
LD	Alkaline phosphatase	FT3
Alkaline phosphatase	Creatinine	Gamma-Glutamyl Transferase (GGT)
GT	Glucose, not fasting	Glucose
Creatinine	Albumin	Hemoglobin
Glucose, not fasting	CEA	Lactate dehydrogenase (LD)
Albumin	NSE (neuron-specific enolase)	Leukocytes
CEA	Pro-GRP	Lymphocytes
NSE (neuron-specific enolase)	C-reactive protein	Magnesium
Pro-GRP		Monocytes
C-reactive protein		Neutrophil granulocytes
		P-FT4
		Phosphate
		Platelets
		Potassium
		Sodium
		Thyroid Stimulating Hormone (TSH)

Study B Endocrinal Status Screening	Study B Endocrinal Status during study
FT4	FT4
FT3	FT3
TSH	TSH
Troponine T	Troponine T (optional)
Testosterone	Testosterone (optional)

Exploratory blood and serum collection
Please see section 6.6.1.

Study procedures and/or SOC treatment performed at other locations.
Study visits can be a burden for the participant, especially with regard to the current pandemic situation. In order not to lose study participants, - study data, and minimize number of protocol deviations, the following options can be used for study visits and procedures related to safety measurements:

- The study visit may be performed digitally by video conference. In case of safety concerns, the participant needs to come to the Center for an additional visit.
- Registration of vital signs may be performed at the local office of the general practitioner
- Blood tests with regard to safety evaluation may be performed at the local hospital or at the office of the general practitioner
- Pregnancy test may be performed at the local hospital or at the office of the general practitioner

The Study Nurse contacts the local hospital or general practitioner in order to collect the result of the necessary study procedures. Process and results are documented in the patients' Electronic Medical Record (DIPS) which then is regarded as source data.

Participants in Study A (glioblastoma) need to come to the center in order to receive Standard of Care (SOC) treatment.

For Study B (brain metastasis) participants, the SOC treatment may be performed at the local hospital. Preferably the first few cycles of treatment are given at the center in order to evaluate how well the treatment is received. After the initial courses, the investigator can together with participant decide if treatment can be continued at local hospital. The Study Nurse contacts local hospital for collection of data and documents treatment in the patients' Electronic Medical Record, which then is regarded as source data.

6.6.3 Withdrawal Visit

Participants who discontinue study treatment for any of the reasons described in section 6.7 will be contacted by the study PI or designee or study nurse as soon as possible for safety follow-up evaluations (section 6.9). At the discretion of the study PI and based on the nature of the discontinuation, a premature withdrawal by the participant from study treatment is not equivalent of a full study withdrawal. If participants refuse to return for these visits or are unable to do so, every effort should be made to contact them or a knowledgeable informant by telephone to obtain follow-up information.

6.6.4 After End of Treatment (Follow-up)

After the last treatment cycle, each participant will meet for a final study visit 14 days after the last dose with IMP to perform a final assessment for any adverse events. Participants at the end of treatment will have post-treatment radiographic and clinical follow-up per study protocol (section 6.5), including assessment of disease status per phone call by the study nurse or designee for overall survival every 2 months until death, withdrawal of consent, or the end of the study monitoring period (2 years from registration), whichever occurs first.

Whereas the participants' involvement in the study will be completed within one year after treatment onset, most patients will likely continue to have relevant anti-cancer treatment and MRI scans under the care of their physician. Consequently, per informed consent, we will assess any new and relevant information equivalent to the radiographic and clinical assessments described in this chapter for one additional year after completion of the study.

6.7 Criteria for Study Treatment Discontinuation

The participants' continuation of study treatment protocol will depend on individual response, evidence of disease progression and tolerance. In the absence of study delays due to adverse event(s), study treatment will continue until one of the following criteria applies:

- Confirmed radiographic and/or clinical progression of neoplastic disease
- Intercurrent illness that prevents continuation of therapy with the IMP
- Intercurrent illness that prevents further MRI exams
- Unacceptable incidental or adverse event(s) of grade 3 (severe) or grade 4 (life-threatening) rendering the participant unacceptable for further therapy with the IMP and/or MRI exams in the judgment of the study PI or treating physician
- Participant demonstrates an inability or unwillingness to comply with study regimen and/or documentation requirements
- Participant decides to withdraw from the study
- General or specific changes in the participants' condition rendering the participant unacceptable for further therapy with the IMP in the judgment of the study PI or treating physician
- Participant is lost to follow-up
- Death

Participants missing one or more of study exams may still continue on study at the discretion of the study PI. Participants who discontinue study treatment for any reason should try to continue post-treatment study protocols. Participants who discontinue for reasons other than progressive disease will have post-treatment follow-up for disease status until disease progression, initiating a non-study cancer treatment, withdrawing consent or becoming lost to follow-up. After documented disease progression each participant will be followed by the study nurse or designee for overall survival every second month by telephone until death, withdrawal of consent, or the end of the study, whichever occurs first.

6.8 Criteria for Study Discontinuation

The study may at any time be discontinued at the discretion of the study PI or the Sponsor in the event of any of the following:

- Occurrence of AEs unknown to date in respect of their nature, severity and duration

- Medical or ethical reasons affecting the continued performance of the trial
- Difficulties in the recruitment of patients
- Lack of adequate sponsor resources

6.9 Procedures for Discontinuation

6.6.1 Patient Discontinuation

Participants have the right to withdraw from the study at any time for any reason. In the case that a participant decides to prematurely withdraw from the study, he or she must be asked if they can still be contacted for further non-treatment assessments, so that a final evaluation can be made with an explanation of why the patient is withdrawing from the study, including assessment of possible adverse events. Although a participant is not obliged to give his or her reason(s) for withdrawing prematurely from a trial, the investigators should make a reasonable effort to ascertain the reason(s), while fully respecting the participants' rights. If available, the reason for removal from protocol therapy, and the date the participant was removed, must be documented in the electronic CRF

If possible, at the last visit of the participant, all assessments of a "Withdrawal visit" will be performed in concordance with study protocol. The investigators are obliged to follow up any significant AE/SAE until the outcome is either recovered or resolved, recovering or resolving, not recovered or not resolved, recovered or resolved with sequelae, fatal or unknown.

6.6.2 Trial Discontinuation

The sponsor and study PI will inform all investigators, the relevant Competent Authorities and Ethics Committees of the termination of the trial along with the reasons for such action. If the study is terminated early on grounds of safety, the Competent Authorities and Ethics Committees will be informed within 15 days.

7 ASSESSMENTS

7.1 Efficacy assessments

Tumor response will be evaluated locally by the investigators according to the procedures described in this chapter. Each participant will be evaluated for CNS disease at pre-therapy baseline exam and thereafter at every relevant visit as shown in section 6.5. This setup will allow us to investigate the radiographic, clinical, safety and tolerability aspects of the IMP. For all biomarker samples, collection information must be captured on the Biomarker Assessment page of the eCRF. Based on the sole decision of the patients' treating physician, the study assessments (primary and secondary endpoints) described in this chapter may at any time be used to evaluate the status of the disease.

All measurements will be recorded in metric notation using a digital measurement tool. The same method of assessment and the same technique should be used to characterize each identified and reported lesion at baseline and during follow-up. Imaging-based evaluation is preferred to evaluation by clinical examination unless the lesion(s) being followed cannot be imaged but are assessable by clinical exam.

All continuous variables will be reported by their mean/median value or a parameter distribution metric (normalized histogram peak height, skewness, kurtosis, location and/or shape of histogram tail)

The following pre-processing steps will apply to all MRI data:

- **Within-exam image analyses** are realigned using normalized mutual information coregistration[104].
- **Between-exam longitudinal analyses** are realigned using a non-linear image coregistration (*nLIC*) pipeline for unbiased matching of MRIs from different exams. Because the tumor mass change with growth or therapy, simply superimposing MRIs from consecutive visits to follow change is not possible due to inherent non-linear voxel displacements. Using our high-res anatomical MRIs, we developed a method for highly accurate within-patient

alignments of images from multiple MR exams by combining robust linear (affine) registration[105] with non-linear image warps. Here, control points in non-tumor, healthy regions with minimal displacement is used to guide the geometry of the warping ('recipe' on how to match two images), where the displacements in the tumor is obtained by finding the best match by minimizing the sum of (**1**) how probable a certain location is and (**2**) how probable the displacement is relative to adjacent deformations. Because small displacements usually return more probably coordinates, the images are slightly smoothened and effectively making the *nLIC* limited to tracking smaller tumor regions at the resolution of our low-res MRIs. The *nLIC* procedure will bypass a well-known hurdle in longitudinal analysis where variations in the patient's positioning introduce up to a 20% intra-reader error[106].

- Image analyses are performed using commercial software nordicICE (NordicNeuorLab AS, Bergen, Norway) and Matlab (MathWorks Inc. Natick, MA, USA), except for the *nLIC* analysis that will also be performed using the free softwares ANTS and FreeSurfer.

7.1.1. Efficacy assessment of the primary endpoints

- **Relative cerebral blood flow** (rCBF) defined by the Zierler's area-height relationship from the perfusion MRI-based tissue-concentration curve of the intravascular contrast agent bolus and normalized to healthy-appearing reference tissue[107,108]. rCBF runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent blood flow and 1 is equivalent of a tumor value equal to healthy-appearing reference. The perfusion MRIs are analyzed using established tracer kinetic models and corrected for contrast agent leakage (from blood-brain-barrier breakdown)[107]. Outlined by a trained neuroradiologist, the region of interest for primary endpoint analysis will constitute a primary **target area** (**PTA**) including any contrast agent enhancing lesion or post-surgical cavity on the MR images (equaling the Gross Total Volume, GTV), as well as a 0.5 centimeter wide area surrounding the GTV, and while at all times within the brain tissue (for **Study A – newly diagnosed glioblastoma** and **study B**). The 0.5-centimeter wide area surrounding the GTV represents the radiological definition of the peri-tumoral zone[79].
- **Solid stress** is derived from MR Elastography by estimating a representative biomarker by tissue stiffness, and normalized to healthy-appearing reference. Tissue stiffness is defined as the magnitude of the shear modulus $|G^*|$, and includes both elastic and viscous information and relates to the distinction between stiff and soft materials [119]. The parameter is normalized to healthy-appearing reference tissue [107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no stiffness. Outlined by a trained neuroradiologist, the region of interest for primary endpoint analysis will constitute a PTA that includes the GTV + a 0.5-centimeter wide area surrounding the GTV, and while at all times within the brain tissue (for **Study A – newly diagnosed glioblastoma** and **study B**). The 0.5- centimeter wide area surrounding the GTV represents the radiological definition of the peri-tumoral zone[79].

7.1.2. Assessments of the secondary endpoints

Outlined by a trained neuroradiologist, the region of interest for the MRI-based secondary endpoint analysis will constitute five separate regions including;

- Any cavity or contrast agent enhancing lesion on the post-contrast MR images (GTV)
- A 0.5 centimeter wide area surrounding the enhancing region or surgical cavity on the post-contrast MR images representing the radiological definition of the peri-tumoral zone[79]
- The surrounding MRI-based FLAIR hyperintensity (without enhancing area) representing intracranial edema
- A 0.5- centimeter wide area surrounding the MRI-based FLAIR hyperintensity representing the peri-edematous zone.
- Healthy brain tissue as identified on the MR images after removal of any tumor and/or edematous regions.

The following functional MRI endpoints will be reported:

- **Perfusion MRIs** are analyzed using established tracer kinetic models and corrected for contrast agent leakage (from blood-brain-barrier breakdown)[107].
    - **Relative cerebral blood volume** (rCBV) defined as area under the MRI-based tissue-concentration curve of the intravascular contrast agent bolus and normalized to healthy-appearing reference tissue[107]. rCBV runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent blood volume.
    - **Mean transit time** (MTT) defined by central volume principle postulating that the level of MRI-based tissue blood volume is equal to its flow multiplied by the time the tracer uses to pass through the tissue[108].

  - MTT runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent MTT.
  - **Extravasation fraction** is defined as the fractional reduction of the MRI-based contrast agent concentration during the conversion from arterial to venous blood[109]. Extraction fraction runs from 0 to 1 where 0 is no exchange and 1 is 100% exchange.
  - **K^{trans}** defined as the volume transfer constant reflecting the efflux rate of gadolinium contrast from blood plasma into the tissue extravascular extracellular space, and used as an indirect measure of capillary permeability[110]. K^{trans} runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no permeability.
- **Vessel architectural imaging**[111] exploits the difference in magnetic sensitivity of gradient-echo and spin-echo MRIs to create unique signal curves that can be presented as parametric plots:
  - **Relative mean vessel caliber** defined as the ratio of the area under the curve of the gradient-echo (total blood volume) to spin-echo (microvessel blood volume) contrast agent MRI perfusion bolus signals[12]. The parameter is normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent vessel caliber.
  - **Relative mean vessel density** defined as the ratio of the area under the curve of the spin-echo (microvessel blood volume) to gradient-echo (total blood volume) contrast agent MRI perfusion bolus signal[12]. The parameter is normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent vessel density.
  - **Vessel size index** (VSI) defined as the relative mean vessel caliber divided by the diffusion coefficient of water in tissue and blood volume[112]. The VSI parameter is normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent VSI.
  - **Corrected vortex area** defined as the area inside an image voxel-wise parametric plot of the dynamic curves of the gradient-echo (total blood volume) to spin-echo (microvessel blood volume) contrast agent MRI perfusion bolus signals[111]. The area is divided by the length of the long axis of the parametric plot (~equivalent to blood volume) to resulting in a 'corrected' vortex area, and normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent vortex area.
  - **Direction map** defined as defined as the propagating direction of the curve of an image voxel-wise parametric plot of the dynamic curves of the gradient-echo (total blood volume) to spin-echo (microvessel blood volume) contrast agent MRI perfusion bolus signals[111]. The parameter is normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 1 (all voxels are counter-clockwise) to 5 (all voxels are clockwise), and where 3 is the equivalent of no apparent direction curve.
- **Diffusion MRIs** will be assessed using Fiber Assignment by Continuous Tracking[113,114]:
  - **Apparent diffusion coefficient** (ADC) defined as the MRI-based diffusion coefficient of the tissue assessed using Stejskal-Tanner diffusion approximation[115]. The parameter is normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent ADC.
  - **Fractional anisotropy (FA)** defined as the normalized variance of the diffusion eigenvalues. The fractional anisotropy reflects differences between an isotropic diffusion and a linear diffusion. The value range is between 0 and 1 (0 = isotropic diffusion, 1 = highly directional).
  - **Restricted spectrum imaging (RSI)** defined as an MRI-based multi-b-value, multi-diffusion time acquisitions that allows for separating the volume fraction and orientation distribution and restricted MRI-based diffusion[116]. The parameter cellular index (CI) from RSI are normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no apparent diffusion.
- **MR Elastography (MRE)** data are estimated based on the external share waves induced in the brain during the MRE acquisition and solved by shear wave propagation analysis[118]. The shear wave is introduced by an external mechanical transducer and produce parameters of:
  - **Tissue viscosity**, defined as the phase angle φ of the complex shear modulus, and describes the dissipative behavior of tissue determined by the cellular network[119]. A higher value represents a more viscous and complex tissue structure. The parameter is normalized to healthy-appearing reference tissue[107] and runs on a dynamic scale from 0 and upwards, and where 0 is the equivalent of no viscosity.

The following neurological performance status endpoints will be reported:

- **Karnofsky Performance Score** (KPS)
  - The KPS runs from 100 to 0 where 100 is perfect health and 0 is death (Appendix D). The KPS is reported in this study as assessed by the treating physician.
- **ECOG performance score**
  - The ECOG performance score runs from 0 to 5, where 0 is perfect health and 5 is death (Appendix D). The ECOG performance score is reported in this study as assessed by the treating physician
- **The Neurologic Assessment in Neuro-Oncology (NANO) scale**
  - The NANO Scale evaluates 9 major domains of neurologic function, with each domain being scored on a range from 0 to 2 or 3. The NANO scale results in a neurological outcome at each assessment compared to baseline:
    - Neurological response is defined as a ≥2 level improvement in at least one domain without worsening in other domains from baseline or best level of function that is not attributable to change in concurrent medications or recovery from a comorbid event.
    - Neurologic stability indicates a score of neurologic function that does not meet criteria for neurologic response, neurologic progression, non-evaluable, or not assessed.
    - Neurologic progression is defined as a ≥2 level worsening from baseline or best level of function within ≥1 domain or worsening to the highest score within ≥1 domain that is felt to be related to underlying tumor progression and not attributable to a comorbid event or change in concurrent medication.
    - Non-evaluable should be selected if it is more likely than not that a factor other than underlying tumor activity contributed to an observed change in neurologic function.
    - Not assessed should be scored if the clinician omits evaluation of that particular domain during his/her examination.
  - The assessment is performed by the treating physician

The following radiographical status endpoints will be reported: (refer section 7.2 for standard radiographic definitions)

- **Size of tumor volume** from contrast agent enhanced regions on post-contrast T1-weighted MRIs, and estimated using a volumetric approach[101] that summarize pathological image voxels.
- **Size of intracranial edema** from T2-weighted FLAIR MRI hyperintensities surrounding any contrast agent enhancing regions as seen on post-contrast MRIs, and estimated using a volumetric approach[101] that summarize pathological image voxels.
- **Occurrence of radionecrosis** as assessed by the neuroradiologist
- **Occurrence of pseudoprogression** as assessed by the neuroradiologist
- **State of disease according to the Response Assessment in Neuro-Oncology (RANO) criteria**:
    - Complete response (CR)
    - Partial response (PR)
    - Stable disease (SD)
    - Progression disease (PD)
  - The RANO criteria will be assessed by a neuroradiologist
- **Best overall radiographic response** as assessed by the neuroradiologist
- **Duration of radiographic response** as assessed by the neuroradiologist

The following clinical status endpoints will be reported:

- **Drug tolerance** by reported SAE's
- **Steroid use** during study will be reported by the treating physician, and consist of
    - Occurrence and type of steroid treatment
    - Total dose of steroid treatment during study
- **Blood pressure** defined as the pressure in the arteries during the contraction of the heart muscle (systolic pressure) over the pressure when the heart muscle is between beats (diastolic pressure).
- **Quality of life** (by EORTC standard) as assessed by the treating physician
- **6-months radiographic progression free survival** of CNS disease as assessed by the treating physician
- **Progression free survival (PFS)** of CNS disease as assessed by the treating physician
- **12- and 24-month survival** as assessed by the treating physician

- **Overall survival (OS)** as assessed by the treating physician

7.2 Radiographic definitions of intracranial disease and progression

For the purposes of this study, participants are re-evaluated for intracranial response per imaging intervals until the participant is taken off, or at the end of, the study. Traditional radiographic response and progression will be evaluated using the modified Response Assessment in Neuro-Oncology (RANO) criteria[99,100].

Radiographic definitions

- **Evaluable non-measurable disease.** Participants who have lesions present at baseline that are evaluable, but on follow-up MR examinations do not meet the definitions of measurable disease will be considered evaluable for non-target disease. The response assessment is based on the presence, absence, or unequivocal progression of the lesions.

- **Measurable disease.** Measurable lesions are defined as those that can be accurately measured in at least one dimension (longest diameter to be recorded) as ≥10 mm with MRI – or – compromise more than 30 image voxels on perfusion MRI to ensure adequate parametric statistical assessments. All tumor measurements must be recorded in millimeters in 2D (or decimal fractions of centimeters) and in cc in 3D.

- **For multifocal intracranial disease**, no more than 3 target lesions should be selected for measurement (>=5mm in diameter in at least one dimension or a total tumor mass of >30 MRI voxels). Target lesions should be selected on the basis of their size (starting with the lesion with the longest diameter or biggest volume), be representative of other lesions and lend themselves to reproducible repeated measurements.

- **Non-measurable disease.** Unidimensional measurable lesions, masses with margins not clearly defined, lesions with maximal diameter < 5mm, or a total tumor mass of < 30 MRI voxels.

Radiographic response criteria

- **Complete Response (CR):** Requires all of the following (*):
  - Complete disappearance of all enhancing measurable and non-measurable disease sustained for at least 4 weeks
  - No new lesions
  - Stable or decreasing steroid dose (the steroid dose at the time of the evaluation scan should be no greater than the dose at the time of the baseline scan)
  - Stable or improved clinical status

** Participants with non-measurable disease cannot have a complete response. The best possible response is stable disease.*

- **Partial Response (PR):** All of the following criteria must be met:
  - Greater ≥50% decrease compared to baseline in the sum of products of perpendicular diameters of all measurable enhancing lesions sustained for at least 4 weeks.
  - No progression of non-measurable disease
  - No new lesions
  - Stable or decreasing steroid dose (the steroid dose at the time of the evaluation scan should be no greater than the dose at the time of the baseline scan)
  - Stable or improved clinical status

- **Progressive Disease (PD):** At least one of the following must be true (*):
  1. A >25% increase in the sum of products of all measurable lesions over smallest sum observed (over baseline if no decrease) using the same techniques used at baseline
  2. Clear worsening of any non-measurable disease*
  3. Appearance of any new enhancing lesions
  4. Clear clinical worsening (unless clearly unrelated to this cancer, e.g. seizures, medication side effects, complications of therapy, cerebrovascular events, infection)
  5. Failure to return for evaluation due to death or deteriorating condition

*** Progression of non-measurable intracranial lesions is defined as one of the follows:**

- A lesion at baseline of </=5mm in diameter that increases to >/= 10mm in diameter
- A lesion at baseline of 6-9 mm in diameter that increases by at least 5 mm in diameter

- **Progression with immunotherapy (Study B only):**

Immune-based therapies are expected to be associated with inflammatory changes that may include edema, and therefore be inaccurately interpreted to represent tumor progression (i.e. pseudoprogression). Therefore, for Study B only, we will define radiographic progressive disease by both (I) assessment of enhancing tumor burden only and (II) additional assessment of T2 or FLAIR changes as outlined in RANO.

- **Stable Disease (SD):** All of the following criteria must be met:
  - Does not qualify for CR, PR or PD
  - All measurable and non-measurable lesions are assessed using techniques used at baseline
  - Stable clinical status

A summary of the RANO response criteria is shown below:

Measurement	**Complete Response**	**Partial Response**	**Stable Disease**	**Progressive Disease**
Post-contrast MRI (T1-weighted)	None	≥ 50% decrease	<50% decrease-<25% increase	≥25% increase*
New Lesion	None	None	None	Present*
Corticosteroids	Stable or decrease	Stable or decrease	Stable or decrease	Stable or increasing
Clinical Status	Stable or improved	Stable or improved	Stable or improved	Decrease*
Requirement for Response	All	All	All	Any*
NA=not applicable #: Progression occurs when any of the criteria with * is present Increase in steroid dose alone will not be taken into account in determining progression in the absence of persistent clinical deterioration				

Evaluation of best overall radiographic response

The best overall response is the best response recorded from the start of the treatment until disease progression/recurrence (taking as reference for progressive disease the smallest measurements recorded since study inclusion). The patient's best response assignment will depend on the achievement of both measurement and confirmation criteria. If a response recorded at one scheduled MRI does not persist at the next regular scheduled MRI, the response will still be recorded based on the prior scan, but will be designated as a non-sustained response. If the response is sustained, i.e. still present on the subsequent MRI at least four weeks later, it will be recorded as a sustained response, lasting until the time of tumor progression. Participants without measurable disease may only achieve SD or PD as their best response.

Duration of radiographic response

- **Duration of overall response:** The duration of overall response is measured from the time measurement criteria are met for CR or PR (whichever is first recorded) until the first date that recurrent or progressive disease is objectively documented (taking as reference for progressive disease the smallest measurements recorded since study inclusion, or death due to any cause. Participants without events reported are censored at the last disease evaluation).

- **Duration of overall complete response:** The duration of overall CR is measured from the time measurement criteria are first met for CR until the first date that progressive disease is objectively documented, or death due to any cause. Participants without events reported are censored at the last disease evaluation.

- **Duration of stable disease:** Stable disease is measured from the start of the treatment until the criteria for progression are met, taking as reference the smallest measurements recorded since the treatment started, including the baseline measurements.

Radiographic pseudo-progression:
The tissue damage caused by anti-cancer treatments and radiotherapy in particular may induce a transient radiographic response that mimics that of PD. If radiologic imaging shows initial PD, participants who are not experiencing significant clinical decline, may be allowed to continue on the study. Patients who have radiographic evidence of further progression after up to three months, or who significantly decline clinically at any time, will be classified as progressive with the date of disease progression back-dated to the first date that the participant met criteria for progression. Among patients on this study with initial radiographic PD, tumor assessment should be repeated regularly in order to confirm PD. If repeat imaging after up to three months confirms progressive disease, then the date of disease progression will be the first date the participant met criteria for progression. In determining whether or not the tumor burden has increased or decreased, investigators should consider all target lesions as well as non-target lesions.

7.3 Safety and Tolerability Assessments

Safety and tolerability assessments will be performed during the clinical evaluation at every visit (section 6.6.2) and concurrent collection of any AEs (section 8) at any time. Significant findings that are present prior to the signing of informed consent must be included in the patient file and on the current medical condition page of the eCRF. For the assessment schedule refer to Flow chart in Section 6.5

7.4 Other assessments of tumor response

Per signed informed consent, collected study data will be stored for up to 15 years or according to applicable regulation and further analyzed to address scientific questions and/or development of biological tests related to the IMP, concomitant therapy and/or cancer. The decision to perform any exploratory biomarker research studies on pseudonymous or anonymized data would be subject to a new scientific and ethical review and based on outcome data from this study or from new scientific findings related to the drug class or disease, as well as reagent and assay availability.

In addition to the biomarkers specified in this chapter, exploratory anonymized biomarker research may be conducted on any radiographic or clinical data collected as part of the study. These studies would extend the search for other potential biomarkers relevant to the effects of the drugs given in combination in this study, and/or prediction of these effects, and/or resistance to the treatment, and/or safety, and these additional investigations would be dependent upon clinical outcome and data availability.

Examples of exploratory analyses include:

- **The RPI signature** (distribution of image voxels) will be compared over time using between-exam longitudinal analyses for treatment response. Here, the magnitude of displacement (per axis value) of the centroid of the image voxels in (x,y) coordinates as presented in the scatter plot (**Figure 2**) will constitute a quantitative measure of treatment chance. Any continuous centroid shift equivalent of a rightwise (x-axis; vascular efficiency) and/or downward (y-axis; solid stress) movement between two exams will constitute a positive value change. Outlined by a trained neuroradiologist, the region of interest for primary endpoint analysis will be the PTA zone[79]. Vascular efficiency is estimated from the MRI perfusion images and defined as the relative mean vessel caliber multiplied by the mean blood flow (refer section 7.1.2). Solid stress is defined from the MR Elastography images as defined as the stiffness from the shear wave modulus G* multiplied by the apparent tissue displacement from the relative strain tensor (refer section 7.1.2). The values of the RPI centroid coordinates from the voxel-vise region of interest are in units relative to the values a brain mask of healthy-appearing gray- and white-matter tissue[121], and with axes ranges of [0%,100%] where 0% represents the lowest patient-specific value across exams and 100% represents the highest value (after a 5% bottom/top outlier removal). A significant change in the treatment group compared to baseline value and/or a non- or a non-treatment group is regarded as a treatment-induced change. A favorable treatment response is a centroid shift in the RPI signature towards [x = 100%, y = 0%] and equivalent of a normal-tissue value.

- **For patients with multiple lesions,** each satellite may represent its own cancerous microenvironment[37,40] and we will assess the benefit of monitoring each satellite separately[100]. In order to measure and track the change in MRI parameters, we will select up to 3 lesions to follow. Some patients may only have 1 metastasis while others will have more so for each patient, we will identify up to 3 lesions (the 3 largest) as our target lesions to follow longitudinally. Within each volumetric lesion we will calculate the median tumor value as well as whole-tumor histogram metrics.
- **The dose distribution maps** from the radiotherapy planning software at the trial site are positioned and realigned on computer tomography (CT) images. Therefore, the geometric positions of the CT scans will be coregistered and realigned to the MRI data using unsupervised mutual information algorithms or similar. In addition, by between-exam longitudinal analyses, smaller brain regions receiving different dose exposures will be identified to reveal why some tumor and normal-appearing brain regions become more damaged than others.
- **MRI-based gradients** of impaired vasculature, solid stress and RPI will be assessed using radial region-grow propagation from the contrast agent enhancing lesion and outwards.

8 SAFETY MONITORING AND REPORTING

The study PI is responsible for the detection and documentation of events meeting the criteria and definition of an adverse event (AE) or serious adverse event (SAE). Each patient will be instructed to contact the study PI immediately should they manifest any signs or symptoms they perceive as serious.

The methods for collection of safety data are described below.

8.1 Definitions

8.1.1 Adverse Event (AE)

An AE is any untoward medical occurrence in a patient administered a pharmaceutical product and which does not necessarily have a causal relationship with this treatment.

An adverse event (AE) can therefore be any unfavorable and unintended sign (including an abnormal laboratory finding), symptom, or disease temporally associated with the use of a medicinal (investigational) product, whether or not related to the medicinal (investigational) product.

The term AE is used to include both serious and non-serious AEs.

Since we are using a licensed drug as IMP with a well know safety profile and in lower dosages than in clinical practice, we want to focus on the interaction between IMP and the standard-of-care treatment in the different cohorts included in this study. These participants will have a lot of AEs related to their disease and the standard-of-care treatment. All AEs will be recorded in the participants medical record and AE Log. AEs with a CTC severity grade 3 or more and all AEs assessed as considered related to the IMP will be recorded in the eCRF and be part of the safety analysis. The reason for this is to assess if the IMP is a significant risk compared to the potential benefit of improved tumor vascularity.

8.1.2 Serious Adverse Event (SAE)

Any untoward medical occurrence that at any dose:

- Results in death
- Is immediately life-threatening

- Requires in-patient hospitalization or prolongation of existing hospitalization
- Results in persistent or significant disability or incapacity
- Is a congenital abnormality or birth defect
- Is an important medical event that may jeopardize the participant or may require medical intervention to prevent one of the outcomes listed above.

Medical and scientific judgment is to be exercised in deciding on the seriousness of a case. Important medical events may not be immediately life-threatening or result in death or hospitalization, but may jeopardize the participant or may require intervention to prevent one of the listed outcomes in the definitions above. In such situations, or in doubtful cases, the case should be considered as serious. Hospitalization for administrative reason (for observation or social reasons) is allowed at the discretion of the study PI and will not qualify as serious unless there is an associated adverse event warranting hospitalization.

A pre-planned hospitalization admission (i.e., elective or scheduled surgery arranged prior to the start of treatment) for pre-existing condition is not considered to be a serious adverse event.

8.1.3 Suspected Unexpected Serious Adverse Reaction (SUSAR)

Suspected Unexpected Serious Adverse Reaction: SAE (see section 8.1.2) that is unexpected as defined in section 8.2 and possibly related to the investigational medicinal product(s).

8.2 Expected Adverse Events

The following side effects have been reported with Cozaar®:

Common (may affect up to 1 in 10 people):

- dizziness,
- low blood pressure (especially after excessive loss of water from the body within blood vessels e.g. in patients with severe heart failure or under treatment with high dose diuretics),
- dose-related orthostatic effects such as lowering of blood pressure appearing when rising from a lying or sitting position,
- debility,
- fatigue,
- hypoglycemia,
- hyperkaliemia,
- changes in kidney function including kidney failure,
- anemia.

Uncommon (may affect up to 1 in 100 people):

- somnolence,
- headache,
- sleep disorders,
- palpitations,
- severe chest pain (angina pectoris),
- shortness of breath (dyspnea),
- abdominal pain,
- obstipation,
- diarrhea,
- nausea,
- vomiting,
- urticaria,
- pruritus,
- rash,

- localized edema,
- cough.

Rare (may affect up to 1 in 1,000 people):
- hypersensitivity,
- angioedema,
- vasculitis including Henoch-Schönlein purpura,
- numbness or tingling sensation (paraesthesia),
- syncope,
- atrial fibrillation,
- stroke,
- hepatitis,
- elevated blood alanine aminotransferase (ALT) levels, usually resolved upon discontinuation of treatment.

Serious but rare (may affects more than 1 in 10,000, but fewer than 1 in 1,000 people):
- A severe allergic reaction (rash, itching, swelling of the face, lips, mouth or throat that may cause difficulty in swallowing or breathing).

Not known (frequency cannot be estimated from available data):
- reduced number of thrombocytes,
- migraine,
- liver function abnormalities,
- muscle and joint pain,
- flu-like symptoms,
- back pain and urinary tract infection,
- photosensitivity,
- rhabdomyolysis,
- impotence,
- pancreatitis,
- hyponatremia,
- depression,
- malaise,
- tinnitus,
- dysgeusia.

Please refer the Summary of Product Characteristics (SmPC) for known Adverse Events on Cozaar® label instructions.
www.legemiddelsok.no (search word 'Cozaar') (In Norwegian)
https://www.medicines.org.uk/emc/product/1004/smpc (In English)

8.3 Disease Progression and/or Recurrence

Disease progression can be considered as a worsening of a patient's condition attributable to the disease for which the Investigational Medicinal Product (IMP) is being studied. It may be an increase in the severity of the disease under study and/or increases in the symptoms of the disease. Expected progression of the disease under the study and/or expected progression of signs and symptoms of the disease under study, unless more severe in intensity or more frequent than expected for the patient's condition should not be reported as an AE. Events which are definitely due to disease progression will not be reported as an AE/SAE. However, if the study PI considers that there was a causal relationship between treatment with IMP or protocol design/procedures and the disease progression/recurrence, then this will be reported as an SAE.

Death due to progressive disease is to be recorded in the eCRF, but not as an SAE.

Any new primary cancer (non-related to the cancer under study) should be considered for reporting as an SAE.

8.4 Time Period for Reporting AE and SAE

The recording of AE and SAEs starts at the day of the first randomization. The standard time period for collecting and recording AE and SAEs for all subjects, both on- or off losartan, in the different diagnosis groups will be as follows:

- **Study A** (patients with recurrent glioblastoma) 168 days + 14 days
- **Study A** (patients with newly diagnosed glioblastoma) 238 days + 14 days
- **Study B** (lung cancer patients with metastases to the brain) 270 days + 14 days]

During the course of the study all AEs and SAEs will be proactively followed up for each patient; events should be followed up to resolution, unless the event is considered by the study PI to be unlikely to resolve due to underlying disease. Every effort should be made to obtain a resolution for all events, even if the events continue after discontinuation/study completion.

8.5 Recording of Adverse Events

All Adverse Events (AE) will be registered in the patient's electronic medical record and on a separate worksheet AE Log. The severity of the AE will be graded according to Common Terminology Criteria for Adverse Events (CTCAE) version 4.0. Adverse events assessed with a severity of grade 3 or more and AEs assessed as related to the IMP will be recorded in the eCRF. The study PI or designee will record the following information in the eCRF:

- The nature of the event(s) will be described by the study PI or designee in precise standard medical terminology (i.e. not necessarily the exact words used by the patient).

- The duration of the event will be described including onset date

- All AEs with an intensity defined as grade 3 or 4 according to CTCAE version 4.0

- All AEs regardless CTCAE grading, but regarded as possible, probable and definite related to the study medication losartan.

- The causal relationship of the event to the study medication will be assessed as one of the following:
    - **Unrelated:** There is not a temporal relationship to investigational product administration (too early, or late, or investigational product not taken), or there is a reasonable causal relationship between non-investigational product, concurrent disease, or circumstance and the AE.
    - **Unlikely:** There is a temporal relationship to investigational product administration, but there is not a reasonable causal relationship between the investigational product and the AE.
    - **Possible:** There is reasonable causal relationship between the investigational product and the AE. Dechallenge information is lacking or unclear.
    - **Probable:** There is a reasonable causal relationship between the investigational product and the AE. The event responds to dechallenge. Rechallenge is not required.
    - **Definite:** There is a reasonable causal relationship between the investigational product and the AE.

- Action taken

- The outcome of the adverse event – whether the event is resolved or still ongoing.

It is important to distinguish between serious and severe AEs. Severity is a measure of intensity whereas seriousness is defined by the criteria in Section 8.1. An AE of severe intensity need not necessarily be considered serious. For example, nausea that persists for several hours may be considered severe nausea, but is not an SAE. On the other hand, a stroke that results in only a limited degree of disability may be considered a mild stroke, but would be an SAE.

Events evaluated as unlikely or unrelated will not be considered for SUSAR reporting.

8.6 Reporting Procedure

8.6.1 AEs and SAEs

All adverse events and serious adverse events will be recorded in the patients' medical records. In addition there will be kept an AE Log per patient. AEs will be recorded in the patient's eCRF as defined in section 8.1.1.

SAEs must be reported within 24 hours after the site has gained knowledge of the SAE. Every SAE must be documented by the study PI or designee using the eCRF form. A SAE Report Form must be completed, signed and sent to the study PI. The initial report shall promptly be followed by detailed, written reports if necessary. The initial and follow-up reports shall identify the trial participants by unique code numbers assigned to the latter. An example of a SAE Report Form is found here:

https://www.norcrin.no/documents/2017/05/vedl-lm-2-7-1-crf-eksempler.pdf/?show_document

The sponsor keeps detailed records of all SAEs reported by the investigators and performs an evaluation with respect to causality and expectedness. SAE reports will be evaluated by the sponsor to see if they affect the risk/benefit ratio associated with the study.

8.6.2 SUSARs

SUSARs will be reported using CIOMS forms located in the Investigator Site File and to the Competent Authority according to national regulation. The following timelines should be followed:

The sponsor will ensure that all relevant information about suspected unexpected serious adverse reactions that are fatal or life-threatening is recorded and reported as soon as possible to the Competent Authority in any case no later than seven (7) days after knowledge by the sponsor of such a case, and that relevant follow-up information is subsequently communicated within an additional eight (8) days.

All other suspected unexpected serious adverse reactions will be reported to the Competent Authority concerned as soon as possible but within a maximum of fifteen (15) days of first knowledge by the sponsor.

8.6.3 Annual Safety Report

Once a year throughout the clinical trial, the sponsor will provide the Competent Authority with an annual safety report. The format will comply with national requirements.

8.6.4 Clinical Study Report

The adverse events and serious adverse events occurring during the study will be discussed in the safety evaluation part of the Clinical Study Report.

8.7 Procedures in Case of Emergency

The study PI is responsible for assuring that there are procedures and expertise available to cope with emergencies during the study.

8.8 Safety Committee (SC)

There will be a Safety Committee consisting of three independent study evaluators:

- Bjørn Henning Grønberg, Chairman SC, consultant oncologist MD, PhD, Department of Oncology, St Olavs Hospital, Norway. E-mail: bjorn.h.gronberg@ntnu.no
- Malin Blomstrand, consultant oncologist MD, PhD, Department of Oncology, Sahlgrenska University Hospital, Gothenburg, Sweden. Email: malin.blomstrand@vgregion.se
- Corina Silvia Rueegg, consultant statistician MSc Department of Biostatistics and Epidemiology, Oslo University Hospital, Norway. E-mail: corrue@ous-hf.no

The SC will evaluate the safety of the intervention when at least six glioblastoma patients (**Study A** – both cohorts summarized) in the 100 mg dose group have reached day 42 (the run-in period). Further meetings will be determined based on the outcome of the run-in interim analysis. The meetings will be documented by Meeting Minutes and a DMC charter and stored in the Investigator Site File as part of the Safety Assessment (refer also section 1.6). The DMC charter will be based on the Norcrin template: https://www.norcrin.no/documents/2017/05/vedl-lm-2-1-3-dmc-charter.docx/

All members of the SC will sign a confidentiality agreement (Appendix F) prior to receiving study information and data.

There will be no stopping of the study due to efficacy.

9 DATA MANAGEMENT AND MONITORING

9.1 Case Report Forms

The electronic Data Capture System (eDCS) used for the eCRF in this study is Viedoc (https://www.viedoc.com/). The setup of the study specific eCRF will be performed by designated staff at Clinical Trial Unit, Data management department at Oslo University Hospital. The eCRF system will be FDA Code of Federal Regulations 21 Part 11 compliant.

The study PI, designee, or study nurse will enter the data required by the protocol into the eCRF. The study PI is responsible for assuring that data entered into the eCRF is complete, accurate, and that entry is performed in a timely manner. The eCRFs should be filled out completely by the Investigator or designee as stated on the delegation of responsibilities form. The signature of the study PI or sub investigator will attest the accuracy of the data on each eCRF. If any assessments are omitted, the reason for such omissions will be noted on the eCRFs. Corrections, with the reason for the corrections will also be recorded.

After database lock, the study PI will receive a digital copy of the participant data for archiving at the trial site.

9.2 Source Data

Source data are all information in original records and certified copies of original records of clinical findings, observations, or other activities in a clinical trial necessary for the reconstruction and evaluation of the trial. Source data are contained in source documents (original records or certified copies).

The medical records for each patient should contain information which is important for the patient's safety and continued care, and to fulfill the requirement that critical study data should be verifiable.

To achieve this, the medical records of each patient should clearly describe at least:

- That the patient is participating in the study, e.g. by including the enrollment number and the study code or other study identification;
- Date when Informed Consent was obtained from the patient and statement that patient received a copy of the signed and dated Informed Consent;

- Results of all assessments confirming a patient's eligibility for the study;
- Diseases (past and current; both the disease studied and others, as relevant);
- Surgical history, as relevant;
- Treatments withdrawn/withheld due to participation in the study;
- Results of assessments performed during the study;
- KPS and/or NANO and/or ECOG WHO performance status assessments conducted as part of the study,
- Treatments given, changes in treatments during the study and the time points for the changes;
- Visits to the clinic / telephone contacts during the study, including those for study purposes only;
- Non-Serious Adverse Events and Serious Adverse Events (if any) including causality assessments;
- Date of, and reason for, discontinuation from study treatment;
- Date of, and reason for, withdrawal from study;
- Date of death and cause of death, if available;
- Additional information according to local regulations and practice.

A source data list will be agreed upon before the onset of the study specifying the source at a module or a variable level.

9.3 Study Monitoring

The investigators will be visited on a regular basis by the Clinical Study Monitor. Details will be provided in a risk-based monitoring plan. The monitor will review the relevant CRFs for accuracy and completeness and will ask the site staff to adjust any discrepancies as required. Sponsor's representatives (e.g. monitors, auditors) and/or competent authorities will be allowed access to source data for source data verification in which case a review of those parts of the hospital records relevant to the study will be required.

9.4 Confidentiality

The study PI shall arrange for the secure retention of the patient identification and the code list. Patient files shall be kept for the maximum period of time permitted by the hospital. The study documentation (CRFs, Site File etc.) shall be retained and stored during the study and for 15 years after study closure or according to applicable regulation. All information concerning the study will be stored in a safe place inaccessible to unauthorized personnel.

9.5 Database management

Only personal health information and corresponding data that is really necessary for the research proposed in the ***ImPRESS-losartan*** proposal will be collected and processed. Digital imaging-based personal health information (radiographic) data from the MRI exam will be collected uniquely for the purpose of this research project using the corresponding clinical MRI systems at the trial site. Relevant electronic non-imaging based personal health information data will be collected from the secure patient information computer systems at the trial site. Other relevant personal health information data (laboratory blood samples for exploratory analyses) will be collected per clinical routine at the trial site and stored according to institutional and national regulations for biobanking. The study PI is responsible for keeping a list of all participants in a dedicated and secure location approved by the trial site.

9.5.1. Electronic CRF and Coding

The Investigator or qualified designee is responsible for recording and verifying the accuracy of subject data. The Data management procedures will be performed in accordance with ICH guidelines.

Data entered into the eCRF will be validated as defined in the data validation plan. Validation includes, but is not limited to, validity checks (e.g. range checks), consistency checks and customized checks (logical checks between variables to ensure that study data are accurately reported) for eCRF data and external data (e.g. laboratory data). A majority of edit checks will be triggered during data entry and will therefore facilitate efficient ‘point of entry’ data cleaning.

Data management personnel will perform both manual eCRF review and review of additional electronic edit checks to ensure that the data are complete, consistent and reasonable.

Manual queries may be added to the system by the clinical data management or study monitor.

All updates to queried data will be made by authorized study centre personnel only and all modifications to the database will be recorded in an audit trail. Once the queries have been resolved, the query form will be signed by the study PI.

Once the full set of eCRFs have been completed and locked, the study PI will authorize database lock and all electronic data will be extracted for statistical analysis according to this protocol. Subsequent changes to the database after database lock will be made only by written agreement approved by the study statistician and the study PI.

Adverse events and medical history will be coded from the verbatim description (Investigator term) using the Medical Dictionary for Regulatory Activities (MedDRA). Prior and concomitant medications and therapies will be coded according to the World Health Organisation drug code.

9.5.2. Anonymization and Pseudonymization of personal health information

- **Anonymous data:** In the context of the ***ImPRESS-losartan*** study, anonymous data is defined as any information from which the person to whom the data relates cannot be identified, whether by the company processing the data or by any other person. Data can only be considered anonymous if re-identification is impossible, meaning that re-identifying an individual must be impossible by any party and by all means likely reasonably to be used for this attempt.

- **Pseudonymous data:** In the context of the ***ImPRESS-losartan*** project, pseudonymous data is defined as any form of de-identification, in which information remains personal data. Pseudonymization is a procedure by which the identifying fields within a data record are replaced by one or more artificial identifiers, or pseudonyms. Pseudonymous data in the ***ImPRESS-losartan*** project will still allow for some form of indirect or remote re-identification by use of a code system that connects the research data (i.e. radiographic images) with a unique participant number to the corresponding personal health information (here identified by the personal identification number and full name).

Any research data at the trial site with exception of the Subject Enrollment and Identification Log will be pseudonymized using participant numbers and otherwise de-identified of any personal health information. The Subject Enrollment and Identification Log will be kept in an electronic database MedInsight OUS with restricted access control.

After a minimum of fifteen years after the ***ImPRESS-losartan*** study end, all research data will be fully anonymized by destruction or deletion of the Subject Enrollment and Identification Log as well as any other form of personal health information that connects the research data with the individual research participant. The study participants are informed of this procedure and the timeline for anonymization of data.

For the intended purpose of making public funded research data available through *open access* and *BioSharing repositories,* anonymized research data at study end will be assigned a new random participant number with no connection to the original study data or personal health information. Anonymized research data may be shared with third-party entities as explained on the informed consent form.

9.5.3. Storage of personal health information:

Personal health information will be stored on secure hospital computer systems (i.e. radiographic images on the hospitals’ picture archiving and communication system, PACS) according to the standard procedures at the trial site as well as on available secure servers dedicated for this purpose only. This includes the servers affiliated with the services for sensitive data (TSD) at the University of Oslo (using the dedicated ***ImPRESS project number 505***).

No personal health information will be accessible to unauthorized personnel.

No personal health information will be transferred outside the secure computer systems at the trial site, nor to any outside institutions. Only pseudonymized data will be shared with collaborators and/or third-parties outside the trial site upon approval. For the purpose of *open access* data, only fully anonymized data is shared.

Pseudonymized research data will only be stored and processed using designated computer systems at the trial site. This includes both laptops and workstations.

9.5.4. Protection of personal health information:

Access to confidential information by an employee affiliated with the study will only be performed on a need-to-know basis and following the participants' signed informed consent.

Study employees will not disclose confidential information to outsiders (including family and friends) or to other employees who do not have a need-to-know.

Study employees will not discuss confidential information in public areas or where others who do not have a need-to-know can overhear the conversation.

Study employees will not make inquires for other personnel who do not have proper authority.

Study employees are responsible for clinical and non-clinical information accessed with the employees' password on a hospital workstation. The employee is also responsible for every action that is made while using the password.

Study employees will not make any unauthorized transmissions, inquiring, modifications, or purging's of data in the hospital computer system. Such unauthorized transitions include, but are not limited to, removing and/or transferring data from the hospital computer system to unauthorized locations outside the hospital system.

Study employees will always lock or log off a hospital workstation prior to leaving it unattended.

Study employees will not share encryption login passwords.

Study employees will not physically remove or transport any personal health information from the trial site, unless such information will be used for the performance of their job duties and in compliance with this policy.

Study employees will ensure that all personal health information whether in paper or electronic format, that is physically removed from the trial site for their job duties is secured and transported in compliance with this policy and referenced policies.

Study employees will not physically remove any personal health information stored in electronic form on laptops or USB devices from the trial site unless the laptop or USB device on which it is stored is in compliance with all applicable trial site encryption policies and standards.

Study employees will not remove personal health information in non-electronic form (whether originals, copies, replications, or reproductions) from the trial site locations, except to transport between such spaces, unless the workforce member has received approval from the study PI, for the purpose of performing work offsite in accordance with the procedures described in this document. Before approving the request, the study PI must be satisfied that the workforce member will implement proper safeguards for protecting the information when physically removed from the trial site locations.

Study employees who transport personal health information in any form, shall take reasonable precautions to safeguard and secure the information at all times.

Study employees will transport the minimum information necessary to perform their duties and so that it is not viewable or accessible to others.

Study employees will not leave personal health information publicly unattended or unsecured at the trial site at any time.

All laptops, tablets, and netbooks, whether owned or managed by the trial site or any trial site entity, or personally owned and managed, and used to process personal health information must be protected and/or encrypted according to institutional guidelines. Study employees will not store personal health information on unencrypted laptops, tablets, or netbooks. Study employees will not connect unprotected and/or unencrypted laptops, tablets, or netbooks to the trial site network if said device is not protected and/or encrypted according to institutional guidelines.

Portable USB drives are only appropriate for the temporary storage and transport of files if necessary. To purchase encrypted portable USB drives, users should contact their local materials management coordinator.

As indicated in the contract of employment and prior to performing any related research activities, Study employees are required to read and abide to all relevant documentation on appropriate use and management of protection health information as described in this document. Violation of these procedures may result in corrective action, up to and including termination of employment and/or suspension and loss of privileges.

Regulatory authorities will be allowed audits at the trial site for source data verification in which a case a review of those parts of the hospital records relevant to the study may be required.

9.5.5. Retention of personal health information:

Personal health information will be stored for no longer than it is required for the processing purposes for which it was collected and for a minimum of <u>fifteen years</u> after the duration of the ***ImPRESS-losartan*** project period.

As a general rule, personal health information and other source data will be kept for the maximum period of time permitted by the trial site in concordance with institutional and Norwegian ethical approvals.

Patients may be withdrawn from the study for general reasons (Helsinki Declaration, Appendix B) or for safety reasons. Patients are free to withdraw from the study at any time without the need to give reasons, and without prejudice to further treatment. Patients may be withdrawn from the study at any time at the discretion of the treating physician or study PI if this is deemed in the best interest of the patient. The patient must be notified immediately of this action and the reason for discontinuation shall be recorded in the registration log.

Within *<u>one year</u>* following the ***ImPRESS-losartan*** study end (31-DEC-2023), anonymized data will be uploaded to the appropriate BioSharing repositories to promote open access. This anonymized data includes general clinical information deemed non-personal health information identifiable (age, weight, gender, diagnosis and any prior treatments), imaging data (MRI with exam intervals only, no dates), corresponding results of any tissue analyses as well as study treatments (with doses and intervals, no dates).

9.5.6. Destruction of personal health information:

After a minimum of fifteen *<u>years after</u>* ***ImPRESS-losartan*** study end, all study-specific personal health information will be irreversibly destroyed by deletion of the code-identifying spreadsheet as well as any other form of personal health information that connects the research data with the individual research participant.

Any residual information that could jeopardize the anonymity of participants will be destroyed after a minimum of fifteen *<u>years after</u>* ***ImPRESS-losartan*** study end. This procedure will be clearly stated on the patient information sheet.

10 STATISTICAL METHODS AND DATA ANALYSIS

The suggested study design is an individual-randomized stepped-wedge, assessor-blinded, multi-dose trial on patients with newly diagnosed or recurrent glioblastoma or brain metastases.

The following section outlines the statistical methods and data analyses. In addition to this outline, a separate statistical analysis plan (SAP) will detail the statistical analyses. The SAP will be signed and dated prior to database lock.

Deviations from the original statistical analysis plan will be described and justified when reporting the findings of the trial, in accordance with the CONSORT statement.

10.1 Determination of Sample Size

The underlying repeated measures mixed model is defined as:

$$\boldsymbol{\triangle P_{ij} = Ai + \mu_{trt} + \mu_j + \epsilon_{ij}}$$

where $\triangle P_{ij}$ is the change from the MRI-based baseline Perfusion value, measured as a percentage of the optimal value (100%) of patient i = 1..*N* at timepoint *j*, A_i is the random intercept of patient *i*, μ_{trt} is the treatment effect , μ_j is the expected P_{ij} at timepoint *j*, and ϵ_{ij} is the residual. Note that we assume the treatment effect is equal between the treatment doses, such that the μ_j denotes the average treatment effect over the three doses. We assume the effect size to be 20%, however, we also assess the power of alternative effect sizes of 5%, 10% and 15%. The variance is difficult to assess, but we assume an overall standard deviation of 15%[89], split with a within- and between patient standard deviation of 10.6%. We used simulations to estimate the sample sizes and corresponding powers separately for each of the three indications.

10.2.1. Simulations

A total of 1000 simulated trials were simulated for each sample size. For each simulation we estimated the parameters of interest (treatment effect) and the p-value from a mixed model. The power is calculated as the proportion of trials with significant treatment effect. For each simulated trial we simulate 6, 12 or 18 patients in each intervention group (step). In the calculations, the doses are pooled to assess the overall treatment effect of losartan.

Sample size per group	Effect size	Total sample size	Power	Estimate
6	5	18	0.187	4.822
6	10	18	0.649	10.214
6	15	18	**0.933**	15.028
6	20	18	**0.998**	20.225
12	5	36	0.353	4.917
12	10	36	**0.921**	10.215
12	15	36	**0.999**	14.973
12	20	36	**1.000**	19.987
15	5	45	**0.437**	5.047
15	10	45	**0.961**	10.070
15	15	45	**1.000**	15.007
15	20	45	**1.000**	19.843
18	5	54	0.498	4.931
18	10	54	**0.979**	10.057
18	15	54	**1.000**	14.968
18	20	54	**1.000**	19.995

** Power calculations shown in bold font are assumed of sufficient magnitude.*

Following the simulations, we see that with an effect size of 20% we have very high power for all sample sizes. With 15 patients in each of the indications, we see that we gain a sufficient power even when the effect size is as low as 10%.

As a result, we need 45 completed patients in each of the two glioblastoma indications, and 36 with brain metastases. To adjust for a possible 15-20% drop-out (non-compliance and mortality), we need to randomize 54 patients in each of the glioblastoma indications and 45 with brain metastases, to a total of 153 patients.

In addition to looking at the primary biomarker endpoint, we want to assess the long-term effect of the intervention on clinical endpoints. The purpose of this long-term follow-up is exploratory (hypothesis generating), as a properly powered hypothesis testing trial on e.g. overall survival would require a lot more patients. However, to have any power to generate hypotheses on clinical endpoints we will randomize half of the patients on 25 mg and 50 mg to either continued or discontinued losartan treatment (same dose), while patients on 100 mg will all continue losartan treatment until day 168 (Study AR) or 238 (Study AN). Following this design, we will end up with a distribution of 18:9:9:18 with no treatment, 25 mg, 50 mg and 100 mg, respectively. The 2:1:1:2 distribution has more power to show a dose-response relationship compared to a standard 1:1:1:1 distribution.

10.2 Randomization

10.2.1 Allocation- sequence generation

Eligible patients will be randomly allocated with equal ratio to each of the treatment groups for patients with glioblastoma (**Study A**), or 1:1:1 ratio to each of the treatment groups for patients with brain metastases (**Study B**). Further will patients in Study A on 25 or 50 mg losartan dose be allocated to either continued or discontinued treatment with the same dose at day 45. The allocation sequence will be computer generated, with one allocation-sequence for each of the three indications. The allocation sequence will detail both the allocation at baseline and at day 45 for Study A. The random allocation-sequence will be stratified within each indication.

Details of block size and allocation sequence generation will be provided in a separate document unavailable to study personnel enrolling patients or assigning patient.

10.2.2 Allocation-procedure to randomize a patient

At the screening visit, patients will be sequentially allocated a screening number. They will be identified by this number until entry into study. As soon as informed consent has been obtained and all inclusion and no exclusion criteria are fulfilled, the patient will be randomized and is allocated a subject number and IMP treatment regime.

The randomization will be performed in a separate randomization module of the Viedoc eCRF. The randomization allocates IMP treatment start time, IMP dose and duration of IMP treatment. For subjects participating in Study A, there will be performed two randomizations; one before treatment start (Cycle 1 – Day 1) and one at Day 40-44 to allocate continued or discontinued treatment. Subjects participating in Study B will only be randomized once, before treatment start (Cycle 1 – Day 1).

The randomization number is equal to the subject number and will replace the screening number.

Each randomization number can only be used once. Patients will not be replaced on study. Enrollment of new patients may be considered until the planned number of evaluable patients is achieved within the indication.

10.2.3 Blinding and emergency unblinding

This is an open label study and therefore no unblinding procedures apply. The treating physician and patient will always be unblinded to treatment allocation. Trial personnel responsible for assessing the radiographic outcomes and the trial statistician will be blinded to treatment allocation.

10.3 Population for Analysis

The following populations will be considered for the analyses:

- Full Analysis Set: Includes all randomized participants, regardless of protocol compliance
- Per-protocol population (PP): Includes all participants who sufficiently comply with the protocol. Criteria for inclusion in the PP population will be specified in the statistical analysis plan prior to database lock.
- Safety population: Includes all participants who have received at least one dose of study medication.

10.4 Planned analyses

There will be a safety interim (run-in) analysis when at least 6 glioblastoma patients in the 100 mg dose group (**Study A** – both cohorts summarized) have reached day 44. See section 8.8 for details.

There will be a futility interim analysis when approximately 50% of the patients have completed the primary endpoint. The aim of this interim analysis is only to assess efficacy with the intention to continue or stop the study based on a futility assessment. Futility is here defined such that the 95% confidence intervals of the estimated treatment effect is within the margin of a relevant effect size. The futility assessment will be detailed in the statistical analysis plan, but could include calculations of conditional power, simulations and analyses of secondary endpoints.

The main statistical analysis is planned when:

- The planned number of patients have been included
- All included patients have either finalized their last assessment or has/is withdrawn according to protocol procedures
- All data have been entered, verified and validated according to the data management plan

Prior to the main statistical analysis, the database will be locked for further entering or altering of data. The statistical analysis will be performed as described in this study protocol.

10.5 Statistical Analysis

This randomized clinical trial aims primarily to describe and estimate efficacy parameters and test pre-specified statistical hypotheses.

10.5.1 Statistical model

The primary variable will be analyzed using a repeated measures linear mixed model with treatment with losartan and dose as primary exploratory variables. There will be no adjustments for baseline characteristics. The dose-response relationship between losartan dose and MRI-based response on the tumor will be assessed by the Multiple Comparison Procedure and Modelling (MCP-Mod)[122]. The methodology incorporates both the planning stage and the analysis stage. In the planning stage, we define a set of plausible candidate dose-response models. For each of these models, optimal statistical tests are defined from which final statistical significance tests are based.

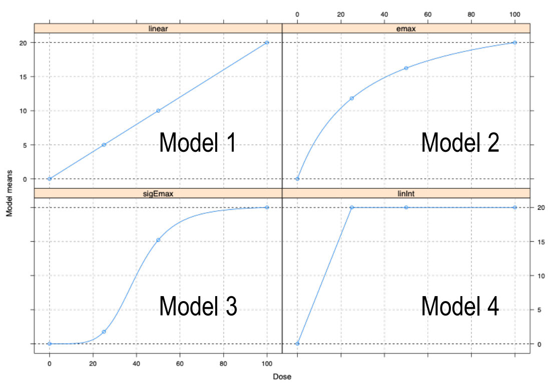

Figure 3: The four proposed models of the Multiple Comparison Procedure and Modelling (MCP-Mod) approach.

Following the trial conduct, the overall dose-response signal is assessed using the linear mixed model on the pre-specified dose-response models, adjusting for multiple testing of multiple candidate models (the MCP step). If a statistically significant dose-response signal has been established in at least one of the tests, we proceed with determining the reference set of significant dose response models by discarding the non-significant models from the initial candidate set. The best model is then selected, and this dose-response model is then used to estimate the dose-response model (the Mod step). Details of the exact MCP-Mod procedure will be given in the SAP but will follow the specifications below.

We assume that the dose-response curves follow one of four different models (**Figure 3**):

1. A linear model
2. An EMax model with an ED50-parameter of 30
3. A sigmoid EMax model with an ED50-parameter of 40 and steepness parameter of 5
4. A step model where the response is the same for all doses.

A two-sided hypothesis test on the 5% level will be used to test the null hypothesis of no dose-response against the alternative hypothesis that the dose-response follows any of the four pre-specified models.

The MCP-Mod methodology will first be applied to the Perfusion primary endpoint. If the MCP-step establishes a statistically significant dose-response signal, the same methodology will be applied to the solid stress primary endpoint. Further dose-response assessments of the clinical endpoints (progression free survival and overall survival) will be also done using the MCP-Mod methodology.

For study B there will only be a test of the null hypothesis of no treatment effect based on the mixed model.

All secondary endpoint variables and exploratory variables will be presented descriptively, and analyzed by, but not limited to, the following methods:

- Continuous secondary variables will be subject to repeated measures mixed models (as for the primary variable), analyses of covariance, or appropriate non-parametric alternatives
- Dichotomous outcome variables will be analyzed using mixed logistic regression models in case of repeated measures, or standard logistic regression models for single dichotomous outcomes.
- Ordinal polytomous outcome variables will be analyzed using cumulative logistic regression models
- Time-to-event variables will be analyzed using Cox proportional hazard models

Unless otherwise specified in the SAP, all statistical considerations will be on the two-sided 5% significance level, and all efficacy assessments will be presented with the appropriate treatment difference measure in addition to the 95% confidence limits.

10.5.2 Adjustments for covariates

Preliminary there will be no adjustments for covariates, but this position may be changed during the trial. The final specification will be made in the SAP.

10.5.3 Handling of drop-outs and missing data

All subjects recruited into the study will be accounted for, including those who did not complete the study along with the reasons for withdrawal.

Missing data in the primary analyses will be handled by the linear mixed model, assuming missing at random (MAR) values. Missing values for secondary analyses will be handled using appropriate methods such as multiple imputation, complete case analyses, last observation carried forward, or worst/best case imputation to reduce bias or, in case bias is unavoidable, conservatively direct the bias. Handling of missing data will be further specified in the SAP. Several handling methods may be applied to the same analyses for assessing the robustness of the results.

10.5.4 Multiple comparisons / multiplicity

Each of the three indications (recurrent and newly diagnosed glioblastoma, brain metastases) will be analyzed separately, and there will be no adjustment for multiple testing between indications. Within each indication analysis, there will be adjustment for multiplicity on the set of candidate dose-response models. Four different adjusted p-values will be presented, and the null hypothesis of no dose-response will be discarded if at least one of the adjusted p-values is below the 5% level. The two primary endpoints will be adjusted for multiple testing by sequentially testing the null hypothesis of no dose-response relationship. First the perfusion primary endpoint will be tested. If statistically significant, the solid stress primary endpoint will be tested. If the perfusion primary endpoint hypothesis test is not rejected, the solid stress primary endpoint hypothesis test will also be regarded as not rejected.

10.5.5 Examination of subgroups

For the primary endpoint:

- **Study A** – both cohorts: IDH gene mutation status (IDH1/2-mutated; yes/no)
- All studies: Steroid use during MRI (yes/no)

For the secondary endpoints:

- **Study A** (both cohorts): IDH gene mutation status (IDH1/2-mutated; yes/no)
- **Study A** (recurrent glioblatoma): temozolomide versus lomustine
- **Study A** (both cohorts): MGMT status (methylated / unmethylated)
- **Study B**: above versus below 50% PDL1 expression

10.5.6 Exploratory analyses

To examine response according to pre-treatment levels of imaging biomarkers, the study sample will be divided retrospectively according to the primary endpoint of response or non-response. Pre-treatment biomarker levels will be summarized descriptively for the two response groups and compared using Wilcoxon rank-sum tests for markers measured on a continuous scale, or Fisher's exact tests for those measured on a categorical scale. Where appropriate, visualization of the relationship between baseline marker levels and the distributions of PFS or OS will employ Kaplan-Meier estimates stratified by biomarker levels. Medians of the time-to-event endpoints will be shown with two-sided 90% confidence intervals; the distributions of PFS and OS will be compared across biomarker strata using the log-rank test. Changes in biomarkers between pre-treatment and progression or treatment discontinuation will be calculated (post-pre) for each patient and summarized descriptively.

11 STUDY MANAGEMENT

11.1 Investigator Delegation Procedure

The study PI is responsible for making and updating a "delegation of tasks" listing all the involved co-workers and their role in the project. He will ensure that appropriate training relevant to the study is given to all of these staff, and that any new information of relevance to the performance of this study is forwarded to the staff involved.

11.2 Protocol Adherence

Investigators ascertain they will apply due diligence to avoid protocol deviations. All significant protocol deviations will be recorded and reported in the Clinical Study Report (CSR).

11.3 Study Amendments

If it is necessary for the study protocol to be amended, the amendment and/or a new version of the study protocol (Amended Protocol) must be notified to and approved by the Competent Authority and the Ethics Committee according to EU and national regulations.

11.4 Audit and Inspections

Authorized representatives of a Competent Authority and Ethics Committee may visit the investigational site to perform inspections, including source data verification. Likewise, the representatives from sponsor may visit the center to perform an audit. The purpose of an audit or inspection is to systematically and independently examine all study-related activities and documents to determine whether these activities were conducted, and data were recorded, analyzed, and accurately reported according to the protocol, Good Clinical Practice (GCP), and any applicable regulatory requirements. The study PI will ensure that the inspectors and auditors will be provided with access to source data/documents.

12 ETHICAL AND REGULATORY REQUIREMENTS

The study will be conducted in accordance with ethical principles that have their origin in the Declaration of Helsinki and are consistent with GCP and applicable regulatory requirements. Registration of patient data will be carried out in accordance with national personal data laws.

12.1 Ethics Committee Approval

The study protocol, including the patient information and informed consent form to be used, must be approved by the regional ethics committee before enrolment of any patients into the study. The study PI is responsible for informing the ethics committee of any serious and unexpected adverse events and/or major amendments to the protocol as per national requirements.

12.2 Other Regulatory Approvals

The protocol will be submitted and approved by the applicable competent authorities before commencement of the study. The protocol will also be registered in www.clinicaltrials.gov before inclusion of the first patient.

12.3 Informed Consent Procedure

The study PI or designee is responsible for giving the patients full and adequate verbal and written information about the nature, purpose, possible risk and benefit of the study. They will be informed as to the strict confidentiality of their patient data, but that their medical records may be reviewed for trial purposes by authorized individuals other than their treating physician.

It will be emphasized that the participation is voluntary and that the patient is allowed to refuse further participation in the protocol whenever she/he wants. This will not prejudice the patient’s subsequent care. Documented informed consent must be obtained for all patients included in the study before they are registered in the study. This will be done in accordance with the national and local regulatory requirements. The study PI is responsible for obtaining signed informed consent.

A copy of the patient information and consent will be given to the patients. The signed and dated patient consent forms will be filed in the Investigator Site File binder and date of consent will be recorded in patient’s electronic medical record.

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12.4 Participant Identification

The study PI is responsible for keeping a list of all patients (who have received study treatment or undergone any study specific procedure) including patient's date of birth and personal number, full names and last known addresses.

The patients will be identified in the eCRFs by subject number, patient initials and date of birth (only at screening).

13 TRIAL SPONSORSHIP AND FINANCING

The study will be sponsored and financed from funds obtained from the following sources:

- *South-Eastern Norway Regional Health Authority Grant 2016102* (PI: Kyrre E. Emblem)
  Period: (YYYY-MM-DD) 2016-01-01 – 2019-12-31
- *The Norwegian Cancer Society Grant 6817564* (PI: Kyrre E. Emblem)
  Period: (YYYY-MM-DD) 2016-01-01 – 2019-12-31
- *South-Eastern Norway Regional Health Authority Grant 2017073* (PI: Kyrre E. Emblem)
  Period: (YYYY-MM-DD) 2017-01-01 – 2020-12-31
- *The Research Council of Norway FRIPRO Grant 261984* (PI: Kyrre E. Emblem)
  Period: (YYYY-MM-DD) 2018-01-01 – 2021-12-31
- *The European Research Council Grant 758657* (PI: Kyrre E. Emblem)
  Period: (YYYY-MM-DD) 2018-01-01 – 2022-12-31
- *The Research Council of Norway BEHANDLING Grant 303249* (PI: Kyrre E. Emblem)
  Period: (YYYY-MM-DD) 2020-04-01 – 2024-03-31
- *The Research Council of Norway Project for Scientific Renewal Grant 325971* (PI: Kyrre E. Emblem)
  Period: (YYYY-MM-DD) 2021-01-12 – 2026-04-01

14 TRIAL INSURANCE

The study PI has insurance coverage for this study through membership of the Drug Liability Association (see http://www.laf.no for more details) as per guidelines of the Directive 2001/20/EC of the European Parliament and of the Council of 4 April 2001 and the Norwegian laws.

15 PUBLICATION POLICY

Upon study completion and finalization of the study report the results of this study will either be submitted for publication and/or posted in a publicly assessable database of clinical study results.

The results of this study will also be submitted to the Competent Authority and the Ethics Committee according to EU and national regulations.

All personnel who have contributed significantly with the planning and performance of the study (Vancouver convention 1988) may be included in the list of authors.

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17 LIST OF APPENDICES

Appendix A: Trial Chart for Study A: *Recurrent glioblastomas*

Appendix B: Trial Chart for Group A: *Newly diagnosed glioblastomas*

Appendix C: Trial Chart for group B: *Brain metastasis*

Appendix D: Performance Status Criteria (ECOG and KPS)

Appendix E: Neurologic Assessment in Neuro-Oncology (NANO) scale

APPENDIX A – TRIAL CHART FOR STUDY A: *RECURRENT GLIOBLASTOMAS*

Study A: Recurrent glioblastomas (AR)	SCREENING			TREATMENT PERIOD													FOLLOW-UP PERIOD			
Cycle (14 day-cycle) **O** = standard of care treatment procedures **X** = study specific procedures	**Screening**	**Within -7 days start chemotherapy**	**-4 days start treatment protocol**	**Cycle 1**	**Cycle 2**	**Cycle 3**	**Cycle 4**	**Cycle 5**	**Cycle 6**	**Cycle 7**	**Cycle 8**	**Cycle 9**	**Cycle 10**	**Cycle 11**	**Cycle 12**	**Cycle 13**	**Cycle 14 Safety FU[4]**	**Cycle 15 - 25**	**12 Months**	**24 Months**
Day in study (Day 1 in every Cycle)				**1**	**15±1d**	**29±1d**	**43±1d**	**57±2d**	**71±4d**	**85±4d**	**99±4d**	**113±4d**	**127±4d**	**141±4d**	**155±4d**	**169±4d**	**183±4d**	**197±7d**	**360±7d**	**720±7d**
Informed Consent	X																			
Inclusion/exclusion criteria	X		(X)[1]																	
Demography and Medical History	X																			
Randomization		X	(X)				X[b]													
Physical exam incl. weight	X		(X)[1]				X			X						X				
NANO + ECOG + KPS score	X		(X)[1]		X		X		X							X				
Vital signs: BP + pulse	X		(X)[1]		X	X	X	X		X		X	X[f]	X[e]		X	X			
Height	X																			
ECG	X		(X)[1]																	
QoL = Modified EORTC QLQ-C30/BN20 (NO)	(X)		X				X									X				
X-MRI: imaging ImPRESS-losartan protocol O-MRI: only RANO evaluation		O+X[2]			X	X			O+ X[2]						O+ X[2]			O-every 3rd Month O-Day 360 ±7d		
(PET-CT - separate ICF)		(X)			(X)															
Bloodtest: CNS New-male/female	(O)		(O)																	
Bloodtest: CNS Follow-up (“KONT”) for Temozolomide	O		(O)	(O)		O		O		O		O		O						
Bloodtest: CNS Follow-up (“KONT”) for Lomustine	O		(O)	(O)		O	O		O	O		O	O		O					
Bloodtest for Exploratory Research			X		X		X			X[2]					X[2]					
Pregnancy test (serum)	(X)		X	X		X		X		X		X		X		X				
Losartan Group AR1_1 (25mg) + AR2_4 (50mg)				X	X	X[a]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]					
Losartan Group AR1_2 (25mg) + AR2_5 (50mg)					X	X[a]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]					
Losartan Group AR1_3 (25mg) + AR2_6 (50mg)						X[a]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]					
Losartan Group AR3_1 (100mg)				X	X	X	X	X	X	X	X	X	X	X	X					
Losartan Group AR3_2 (100mg)					X	X	X	X	X	X	X	X	X	X	X					
Losartan Group AR3_3 (100mg)						X	X	X	X	X	X	X	X	X	X					
Chemo:Temozolomide tablets				O		O		O		O		O		O						
Chemo: Lomustine tablets				O			O			O			O							
Concommitant medication	X		(X)[1]			X	X	X		X		X	X[f]	X[e]		X				

Study A: Recurrent glioblastomas (AR)	SCREENING			TREATMENT PERIOD													FOLLOW-UP PERIOD			
Cycle (14 day-cycle) **O** = standard of care treatment procedures **X** = study specific procedures	**Screening**	**Within -7 days start chemotherapy**	**-4 days start treatment protocol**	**Cycle 1**	**Cycle 2**	**Cycle 3**	**Cycle 4**	**Cycle 5**	**Cycle 6**	**Cycle 7**	**Cycle 8**	**Cycle 9**	**Cycle 10**	**Cycle 11**	**Cycle 12**	**Cycle 13**	**Cycle 14 Safety FU[4]**	**Cycle 15 - 25**	**12 Months**	**24 Months**
Day in study (Day 1 in every Cycle)				**1**	**15±1d**	**29±1d**	**43±1d**	**57±2d**	**71±4d**	**85±4d**	**99±4d**	**113±4d**	**127±4d**	**141±4d**	**155±4d**	**169±4d**	**183±4d**	**197±7d**	**360±7d**	**720±7d**
Steroids dosage	(X)	X			X	X	X	X		X		X	X[f]	X[e]		X				
AE/SAE reporting			X			X	X[f]	X[e]		X		X	X[f]	X[e]		X	X			
Survival assessment																X	X[c]	X[3]	X[3]	X[3]

1) Procedures performed within 14 days before study treatment start (C1-D1), may serve as baseline.
2) Screening window of -8 days and for the other visits a window of -8 and +1 day(s), preferable in the second week of the cycle.
For subjects randomized to extended treatment, it is important that the procedures in Cycle 12 are performed when the subject is still on losartan.
3) Survival Assessment: every 2 months (± 7 days)
4) In case of withdrawal from study, same procedures as Safety FU must be performed in addition to recording the reason for withdrawal

a) Day 29-44
b) Day 40-44
c) Day 180 ± 7 days
d) Randomized to either treatment discontinuation or extended treatment
e) Only for subjects on temozolomide
f) Only for subjects on lomustine

APPENDIX B – TRIAL CHART FOR GROUP A: *NEWLY DIAGNOSED GLIOBLASTOMAS*

Study A: Newly diagnosed glioblastomas (AN)	**SCREENING**			**TREATMENT PERIOD**																		**FOLLOW-UP PERIOD**			
Cycle (14 day-cycle) **O** = standard treatment procedures **X** = study specific procedures	**1-3 weeks after surgery**	**-7 days start treatment protocol**	**-4 days start treatment protocol**	**Cycle 1**	**Cycle 2**	**Cycle 3**	**Cycle 4**	**Cycle 5**	**Cycle 6**	**Cycle 7**	**Cycle 8**	**Cycle 9**	**Cycle 10**	**Cycle 11**	**Cycle 12**	**Cycle 13**	**Cycle 14**	**Cycle 15**	**Cycle 16**	**Cycle 17**	**Cycle 18**	**Cycle 19 Safety FU[4]**	**Cycle 20 - 25**	**12 Months**	**24 Months**
Day in study (Day 1 in every Cycle)				**1**	**15±1d**	**29±1d**	**43±1d**	**57±2d**	**71±4d**	**85±4d**	**99±4d**	**113±4d**	**127±4d**	**141±4d**	**155±4d**	**169±4d**	**183±4d**	**197±4d**	**211±4d**	**225±4d**	**239±4d**	**253±4d**	**267±7d**	**360±7d**	**720±7d**
Informed Consent	X																								
Inclusion/exclusion criteria	X		(X)[1]																						
Demography and Medical History	X																								
Randomization		X	(X)				X[b]																		
Physical exam including weight	X		(X)[1]				X								X						X				
NANO + ECOG + KPS score	X		(X)[1]		X		X		X						X						X				
Vital signs: BP + puls	X		(X)[1]		X	X	X		X		X		X		X		X		X		X	X			
Height	X		(X)[1]																						
ECG	X		(X)																						
QoL = EORTC QLQ-C30/BN20 (NO)	(X)		X				X								X						X				
X-MRI: imaging ImPRESS-losartan prot. O-MRI: only RANO evaluation		X[2]			X	X	X							O+ X[2]						O+ X[2]			O-every 3rd Month O-Day 360 ±7d		
Bloodtest: CNS New-male/female		O	(O)																						
Bloodtest: CNS Follow-up ("KONT")			(O)	O	O	O	O		O		O		O		O		O		O		O				
Bloodtest for exploratory research			X		X		X													X[2]					
Pregnancy test (blood)	(X)		X	X		X		X		X		X		X		X		X		X		(X)			
Losartan Group AN1_1 (25mg) + AN2_4 (50mg)				X	X	X[a]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]					
Losartan Group AN1_2 (25mg) + AN2_5 (50mg)					X	X[a]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]					
Losartan Group AN1_3 (25mg) + AN2_6 (50mg)						X[a]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]	(X)[d]					
Losartan Group AN3_1 (100mg)				X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
Losartan Group AN3_2 (100mg)					X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
Losartan Group AN3_3 (100mg)						X	X	X	X	X	X	X	X	X	X	X	X	X	X	X					
Adjuvant Chemo:Temozolomide tablets				O	O	O			O		O		O		O		O		O						
Radiation Therapy				O	O	O	(O)																		

Study A: Newly diagnosed glioblastomas (AN)	**SCREENING**			**TREATMENT PERIOD**																		**FOLLOW-UP PERIOD**			
Cycle (14 day-cycle) **O** = standard treatment procedures **X** = study specific procedures	**1-3 weeks after surgery**	**-7 days start treatment protocol**	**-4 days start treatment protocol**	**Cycle 1**	**Cycle 2**	**Cycle 3**	**Cycle 4**	**Cycle 5**	**Cycle 6**	**Cycle 7**	**Cycle 8**	**Cycle 9**	**Cycle 10**	**Cycle 11**	**Cycle 12**	**Cycle 13**	**Cycle 14**	**Cycle 15**	**Cycle 16**	**Cycle 17**	**Cycle 18**	**Cycle 19 Safety FU[4]**	**Cycle 20 - 25**	**12 Months**	**24 Months**
Day in study (Day 1 in every Cycle)				**1**	**15+1d**	**29+1d**	**43+1d**	**57+2d**	**71+4d**	**85+4d**	**99+4d**	**113+4d**	**127+4d**	**141+4d**	**155+4d**	**169+4d**	**183+4d**	**197+4d**	**211+4d**	**225+4d**	**239+4d**	**253+4d**	**267+7d**	**360+7d**	**720+7d**
Concommitant medication	X		(X)[1]		X	X	X		X		X		X		X		X		X		X				
Steroids dosage	(X)	X			X	X	X		X		X		X		X	X	X		X		X				
AE/SAE reporting			X		X	X	X		X		X		X		X		X		X		X	X			
Survival assessment																	X[c]						X[3]	X[3]	X[3]

1) Procedures performed within 14 days before study treatment start (C1-D1), may serve as baseline.
2) Screening window of -8 days and for the other visits a window of -8 and +1 day(s), preferable in the second week of the cycle.
For subjects randomized to extended treatment, it is important that the procedures in Cycle 17 are performed when the subject is still on losartan.
3) Survival Assessment: every 2 months (± 7 days)
4) In case of withdrawal from study, same procedures as Safety FU must be performed in addition to recording the reason for withdrawal

a) Day 29-44
b) Day 40-44
c) Day 180 ± 7 days
d) Randomized to either treatment discontinuation or extended treatment

APPENDIX C – TRIAL CHART FOR GROUP B: *BRAIN METASTASIS*

Study B: Brain Metastasis (BM)	SCREENING			TREATMENT PERIOD									FOLLOW-UP		
Cycle (90 day-cycle) **O** = standard treatment procedures **X** = study specific procedures	**-21days**	**-7 days**	**-4 days**	**Cycle 1**			**Cycle 2**			**Cycle 3**			**Safety FU[9]**	**12 Months**	**24 Months**
Day in Study Cycle				**Day 1 - 90**			**Day 91 - 180**			**Day 181 - 270**					
				1		**90 -7d**	**91**		**180 -7d**	**181**		**270 -7d**	**284 ±4d**	**360 ±7d**	**720 ±7d**
Informed Consent	X														
Inclusion/exclusion criteria	X		(X)[1]												
Demography, med. history, prior treatment	X		(X)[1]												
Randomization		X	(X)[1]												
Physical exam including weight	X		(X)[1]			X[3]			X[3]			X[5]			
NANO + ECOG + KPS score	X		(X)[1]			X[3]			X[3]			X[5]		X[6]	
Vital signs: BP + pulse	X		(X)[1]	X[2]		X[3]	X[2]		X[3]	X[2]		X[5]	X	X[6]	
Height	X		(X)[1]												
ECG	X		(X)[1]												
QoL= modified EORTC QLQ-C30/BN20 (NO)	(X)		X			X[3]			X[3]			X[5]		X[6]	
MRI: imaging protocol ImPRESS-losartan		O+X				O+X[3+10]			O+X[3+10]			O+X[10]		O[6]	O[7]
Bloodtests Protocol: LUNG NEW	(O)		(O)												
Bloodtests Protocol: LUNG Follow-up («KONT»)	O		(O)	O[2]			O[2]			O[2]		O	O	O	
Bloodtests Protocol: IMMUN 1 and 2[4]	(O)		(O)	(O)[2]			(O)[2]			(O)[2]		(O)	(O)	(O)	
Bloodtests: Endocrine status[11]	(O)		O			O			O			O	(O)	(O)	
Bloodtest for Exploratory Research			X			X[3]			X[3]			X[5]		X[6]	
Pregnancy test (blood)	(X)[1]		X	X[2]			X[2]			X[2]			(X)		
Losartan Group BM1 (50mg)				X	X	X	X	X	X	X	X	X			
Losartan Group BM2 (50mg)							X	X	X	X	X	X			
Losartan Group BM3 (50mg)										X	X	X			
SOC Chemo- and/or Immunotherapy				O – As per Standard of Care according to National Guidelines											
Steroids dosage	(X)	X				X[3]			X[3]			X[5]		X[6]	
Concommitant medication	X		(X)[1]				X			X		X[5]		X[6]	
AE/SAE reporting			X	X[2]			X[2]			X[2]		X[5]	X		
Survival assessment									X[6]			X[6]		X[8]	X[8]

1) Procedures performed within 14 days before study treatment start (C1-D1), may serve as baseline.
2) Every 3rd week in connection with i.v. chemo-/immunotherapy
3) -7 days and for those subjects not started on losartan the procedures have to be performed before IMP treatment start
4) Only for subject on immunotherapy treatment
5) -7 days: Procedures must be performed while the subject is still on losartan
6) ± 7days
7) MR Caput every 3rd month
8) Survival Assessment: every 2 months (+ 7 days)
9) In case of withdrawal from study, same procedures as Safety FU must be performed in addition to recording the reason for withdrawal
10) -14 days
11) Endocrine status: FT3, FT4, and TSH. Testosterone and Troponine T mandatory at screening and optional during treatment

APPENDIX D - PERFORMANCE STATUS CRITERIA (ECOG AND KPS)

ECOG Performance Status Scale		Karnofsky Performance Scale	
Grade	Descriptions	Percent	Description
0	Normal activity. Fully active, able to carry on all pre-disease performance without restriction.	100	Normal, no complaints, no evidence of disease.
		90	Able to carry on normal activity; minor signs or symptoms of disease.
1	Symptoms, but ambulatory. Restricted in physically strenuous activity, but ambulatory and able to carry out work of a light or sedentary nature (*e.g.*, light housework, office work).	80	Normal activity with effort; some signs or symptoms of disease.
		70	Cares for self, unable to carry on normal activity or to do active work.
2	In bed <50% of the time. Ambulatory and capable of all self-care, but unable to carry out any work activities. Up and about more than 50% of waking hours.	60	Requires occasional assistance, but is able to care for most of his/her needs.
		50	Requires considerable assistance and frequent medical care.
3	In bed >50% of the time. Capable of only limited self-care, confined to bed or chair more than 50% of waking hours.	40	Disabled, requires special care and assistance.
		30	Severely disabled, hospitalization indicated. Death not imminent.
4	100% bedridden. Completely disabled. Cannot carry on any self-care. Totally confined to bed or chair.	20	Very sick, hospitalization indicated. Death not imminent.
		10	Moribund, fatal processes progressing rapidly.
5	Dead.	0	Dead.

APPENDIX E - NEUROLOGIC ASSESSMENT IN NEURO-ONCOLOGY (NANO) SCALE

Scoring assessment is based on direct observation and testing performed during clinical evaluation and is not based on historical information or reported symptoms. Please check one answer per domain. Please check "Not assessed" if testing for that domain is not done. Please check "Not evaluable" if a given domain cannot be scored accurately because of preexisting conditions, comorbid events, or concurrent medications.

Patient identifier: ______________________________

Date assessment performed (day/month/year): ______________________________

Study time point (ie, baseline, cycle 1, day 1, etc): ______________________________

Assessment performed by (please print name): ______________________________

Domains | **Key Considerations**

Gait

- Walking is ideally assessed by at least 10 steps

0 ☐ Normal
1 ☐ Abnormal but walks without assistance
2 ☐ Abnormal and requires assistance (companion, cane, walker, etc.)
3 ☐ Unable to walk
☐ Not assessed
☐ Not evaluable

Strength

- Test each limb separately
- Recommend assess proximal (above knee or elbow) and distal (below knee or elbow) major muscle groups
- Score should reflect worst performing area
- Patients with baseline level 3 function in one major muscle group/limb can be scored based on assessment of other major muscle groups/limb

0 ☐ Normal
1 ☐ Movement present but decreased against resistance
2 ☐ Movement present but none against resistance
3 ☐ No movement
☐ Not assessed
☐ Not evaluable

1

Ataxia (Upper Extremity)

0 ☐ Able to finger-to-nose touch without difficulty
1 ☐ Able to finger-to-nose touch but difficult
2 ☐ Unable to finger-to-nose touch
☐ Not assessed
☐ Not evaluable

- Nonevaluable if strength is compromised
- Trunk/lower extremities assessed by gait domain
- Particularly important for patients with brainstem and cerebellar tumors
- Score based on best response of at least 3 attempts

Sensation

0 ☐ Normal
1 ☐ Decreased but aware of sensory modality
2 ☐ Unaware of sensory modality
☐ Not assessed
☐ Not evaluable

- Recommend evaluating major body areas separately (face, limbs, and trunk)
- Score should reflect worst performing area
- Sensory modality includes but not limited to light touch, pinprick, temperature, and proprioception
- Patients with baseline level 2 function in one major body area can be scored based on assessment of other major body areas

Visual Fields

0 ☐ Normal
1 ☐ Inconsistent or equivocal partial hemianopsia (≥quadrantanopsia)
2 ☐ Consistent or unequivocal partial hemianopsia (≥quadrantanopsia)
3 ☐ Complete hemianopsia
☐ Not assessed
☐ Not evaluable

- Patients who require corrective lenses should be evaluated while wearing corrective lenses
- Each eye should be evaluated, and score should reflect the worst performing eye

Facial Strength

0 ☐ Normal
1 ☐ Mild/moderate weakness
2 ☐ Severe facial weakness
☐ Not assessed
☐ Not evaluable

- Particularly important for brainstem tumors
- Weakness includes nasolabial fold flattening, asymmetric smile, and difficulty elevating eyebrows

2

Language

0 ☐ Normal

1 ☐ Abnormal but easily conveys meaning to examiner

2 ☐ Abnormal and difficulty conveying meaning to examiner

3 ☐ Abnormal; if verbal, unable to convey meaning to examiner; OR nonverbal (mute/global aphasia)

☐ Not assessed

☐ Not evaluable

- Assess based on spoken speech; nonverbal cues or writing should not be included
- **Level 1**: Includes word-finding difficulty; few paraphasic errors/neologisms/word substitutions; but able to form sentences (full/broken)
- **Level 2**: Includes inability to form sentences (<4 words per phrase/sentence); limited word output; fluent but "empty" speech

Level of Consciousness

0 ☐ Normal

1 ☐ Drowsy (easily arousable)

2 ☐ Somnolent (difficult to arouse)

3 ☐ Unarousable/coma

☐ Not assessed

☐ Not evaluable

- None

Behavior

0 ☐ Normal

1 ☐ Mild/moderate alteration

2 ☐ Severe alteration

☐ Not assessed

☐ Not evaluable

- Particularly important for frontal lobe tumors
- Alteration includes but is not limited to apathy, disinhibition, and confusion
- Consider subclinical seizures for significant alteration

3